

Models with non-constant rates

Variability in evolution, competitive reversal, and the two phases of infection

Adam Aldridge
University of Leeds

Contents

1	Models with non-constant rates	3
1.1	Introduction	3
1.1.1	Notation	5
1.2	Logistic speciation	7
1.2.1	Variation, and what suppresses it	7
1.2.2	The push of the past	7
1.2.3	The model	7
1.2.4	The marginal law	8
1.2.5	The mean, directly	9
1.2.6	The limit law	9
1.2.7	The other rate laws	10
1.3	Competitive reversal: the Pimentel model	12
1.3.1	The experiment	12
1.3.2	The model	12
1.3.3	Results	14
1.3.4	The moving critical point	15
1.3.5	A spatial version	16
1.3.6	Coupling speciation to competition	16
1.4	Rise, run, ruin, rejuvenation	18
1.4.1	The cycle the model is for	18
1.4.2	The model	19
1.4.3	What the simulations show	19
1.5	The birth-plague model	21
1.5.1	The model	21
1.5.2	The vitality-coupled variant	21
1.6	Public good accumulation	23
1.6.1	Where the idea comes from	23
1.6.2	The model	24
1.6.3	The minimum founding cohort	24
1.7	The two phases of infection	26
1.7.1	The biology, and the shape it dictates	26
1.7.2	Phase composition	26
1.7.3	The host-to-host layer	27
1.7.4	The extinction probability, derived	28
1.7.5	Other compositions	28
1.8	Discussion	30

1.8.1	What the collection shows	30
1.8.2	Limitations	30
1.8.3	Open problems	31
1.8.4	Connections	31
1.A	Model summary	32
1.B	Figure and simulation record	32
1.C	The inhomogeneous Yule generating function	33
1.C.1	The generating function	33
1.C.2	The distribution	34

Chapter 1

Models with non-constant rates

1.1 Introduction

Every model carried through to a result in the preceding chapters holds its rates fixed. A replicative unit is born at rate λ , dies at rate μ and triggers catastrophe at rate δ , and those three numbers are the same at the end of the process as at the beginning. That assumption is what makes the generating functions close, and it buys a great deal: the roots a and b , the burst-size law, the flooding criterion and the removal budget. It is also the first thing an ecologist or an epidemiologist will question, because in almost every setting one might want to describe, a rate is not a constant — resources run out, competitors adapt, and a host mounts an immune response.

This chapter asks what happens when a rate is allowed to move. The models collected here were built at different times and stand at different stages of completion, but they sort by one structural feature: *what moves the rate*. Four mechanisms recur.

Exhaust, no inflow

Remaining room $K - N(t)$ is spent and never returned. The integrated rate $\mathcal{B}(t)$ converges, and S_t has a proper limit law.

logistic speciation

Exhaust, with inflow

Potential V is consumed per capita and restocked at rate ϕ . The process cycles rather than settling.

rise–run–ruin–rejuvenation; vitality

Moving critical point

Resistance update drags the state of no preferred direction after the leader, and so restores criticality.

Pimentel competition

Rupture as a join

The intracellular phase ends at rupture, rate $\delta_1 X_t$. The burst-size law integrates that time out, so the phases compose.

two phases of infection

Figure 1.1: Four ways a rate can move. Remaining room that is spent and never returned yields a proper limit law; a potential that is restocked from outside yields a cycle; resistance that drags a critical point yields reversal; a rupture at rate $\delta_1 X_t$ yields two phases that compose, because the burst-size law has already integrated the random time out.

Where a rate is driven by the process’s own state, the behaviour turns on whether the feedback

exhausts a quantity or replenishes it. Speciation under logistic growth consumes room that is never returned, the integrated rate converges, and the species count has a proper limit law. An accumulating potential, restocked at a constant inflow, instead yields a cycle. Resistance in the Pimentel model does something else again: it acts not on a level but on a *critical point*, dragging the state of no preferred direction after the leading population and so restoring criticality. In the last model the rate jumps at a single instant rather than bending continuously: the intracellular phase ends at the rupture of the host cell, which the process itself brings on at rate $\delta_1 X_t$, and the burst-size law has already integrated that time out. That is why the two phases compose. The ecological models occupy the first half of the chapter; a reader who wants only the *Y. pestis* material will find it in [sections 1.6](#) and [1.7](#).

Four results carry the finished mathematics, and each is stated here so that the derivations may be read knowing where they go.

- For a pure birth process from one founder with per-capita rate $\beta(t)$, the species count S_t is geometric on $\{1, 2, \dots\}$ with parameter $e^{-\mathcal{B}(t)}$, where $\mathcal{B}(t) = \int_0^t \beta(u) du$ ([proposition 1.2](#)). The rate law enters the distribution only through its integral.
- Under the logistic rate the integral converges, so the process has a proper limit law and not merely a finite late-time mean: S_∞ is geometric with mean $(K/N_0)^{\sigma K/\varrho}$ ([corollary 1.3](#)), depending on the parameters through exactly those two dimensionless groups.
- In the Pimentel competition model the state at which the chain is equally likely to step up as down is available in closed form ([\(1.20\)](#)), and it is unstable. Two trajectories started from the same state diverge exactly when they are near it.
- Composing an intracellular birth–death–catastrophe process with an extracellular birth–death process gives the host-to-host mean offspring $\mathbb{E}(Z)$ in closed form ([\(1.38\)](#)), as an explicit function of the phase-1 catastrophe rate.

The Yule process, its geometric marginal and the method of characteristics are the mathematical introduction's; the logistic speciation model itself, with its mean, is already a worked example there. The birth–death–catastrophe process, the roots a and b and the burst-size law are the chapter on the birth–death–catastrophe process's and the chapter on its distribution theory's. None of that is re-derived here. What this chapter adds is the distribution under a moving speciation rate, the moving critical point of the Pimentel chain, the two-phase extinction probability, and the remaining specifications, stated precisely enough to be taken up where they stop.

§	Model	What moves the rate	Status
1.2	Logistic speciation	remaining capacity $K - N(t)$	closed form
1.3	Pimentel competition	evolving resistance, itself driven by the state	closed-form critical point; simulated
1.3.6	Pimentel+	fitness driven by dominance, reduced by species count	specified
1.4	Rise, run, ruin, rejuvenation	potential V ; catastrophe by scarcity X_t/V	simulated
1.5	Birth–plague	loss channel driven by the coupled component Y_t	simulated
1.6	Public good	$\lambda(t) = \lambda_0 + \alpha P(t)$, P made by the population	K^* computed
1.7	Two phases of infection	switch at the rupture t_c ; decaying $\mu_2(t)$	closed form

Table 1.1: The seven models, and the mechanism by which a rate moves in each. The last column is the shape of the chapter: [sections 1.2](#) and [1.7](#) are its two mathematical ends; the rows between them halt at a closed-form fragment, a simulation or a specification. The birth–plague model is carried in two variants, so the count of distinct specifications is eight.

1.1.1 Notation

Symbols are reused across models but never within one section, and never against a thesis-wide meaning. The table records the assignment.

§	Symbol	Meaning
all	λ, μ, δ	birth, death and catastrophe rates, as in the chapter on the birth–death–catastrophe process
	a, b	roots of the characteristic quadratic, $b < 1 < a$ (??)
	$\mathcal{K}$	burst size of a birth–death–catastrophe process
1.2	S_t	species count, $S_0 = 1$
	$N(t), K, N_0$	total abundance, carrying capacity, initial abundance
	ϱ	logistic growth rate
	σ	speciation coefficient; $\beta(t) = \sigma(K - N(t))$
	ξ	the constant $K/N_0 - 1$
	$\beta(t)$	speciation rate per species; β is this and nothing else
	$\mathcal{B}(t)$	integrated rate $\int_0^t \beta(u) du$; η is a constant floor added to β in two of the alternative laws
1.3, 1.3.6	A_t, B_t	counts of the two competing populations, $A_t + B_t = N$
	N	capacity of the enclosure, an integer
	R_a, R_b	resistance of each population
	p_a, p_b	blocking probabilities, at most one non-zero
	ε, α	evolution rate (the step size of the resistance update) and the exponent sharpening that drive near a boundary
	i_c	the critical state (1.20)
	f_a, f_b	fitness, the Pimentel+ recasting of R_a, R_b
	$S_t^{(a)}, S_t^{(b)}$	per-population species counts; ζ couples dominance to speciation and γ penalises fitness per species
1.4	$X_t, V(t)$	count of replicative agencies, and accumulated potential
	ϕ, c, ω	potential inflow, potential consumed per agency, revival rate out of the empty state
1.5	X_t, Y_t	uninfected and infected counts
	χ	contact-infection coefficient
	$V(t), \phi$	vitality and its inflow rate, in the coupled variant only
1.6	$X_t, P(t)$	intracellular <i>Y. pestis</i> count, and public-good concentration
	λ_0, α	base birth rate and public-good advantage
	g, c, K^*	production, clearance, and the minimum founding cohort (1.31)
1.7	$\lambda_1, \mu_1, \delta_1$	phase-1 rates, of the intracellular birth–death–catastrophe process
	λ_2, μ_2	phase-2 rates, of the extracellular birth–death process
	t_c	time of the switch between phases
	Z, p_0	offspring count of the host-to-host process, and the probability that the whole within-host process dies out

Table 1.2: Notation, organised by model. The rates λ , μ and δ , the roots a and b and the burst size $\mathcal{K}$ carry their thesis-wide meanings everywhere in this chapter and are never redefined. Where one letter serves two models — α as an exponent in [sections 1.3](#) and [1.3.6](#) and as a coupling coefficient in [section 1.6](#), V as potential in [section 1.4](#) and as vitality in [section 1.5](#), c as a consumption rate in both, N as an abundance in [section 1.2](#) and as an integer capacity in [section 1.3](#) — the two uses are separated by this table and never occur in the same section.

The ecological models occupy [sections 1.2, 1.3, 1.3.6, 1.4](#) and [1.5](#). The *Y. pestis* models close the chapter: [section 1.6](#) asks for the founding cohort that would make an accumulating public good supercritical, and [section 1.7](#) composes the two phases of plague infection and obtains $\mathbb{E}(Z)$ in closed form. [Section 1.8](#) gathers limitations and the open questions that remain; [section 1.A](#) summarises all seven models on one page, [section 1.B](#) records how the figures were produced, and [section 1.C](#) gives the generating-function proof of [proposition 1.2](#).

1.2 Logistic speciation

1.2.1 Variation, and what suppresses it

The premise of the model is Fisher's: variation is in constant supply, and what limits the number of distinct types is not the rate at which they arise but the severity of the conditions under which they must persist [1]. "To maintain a stationary variance, fresh mutations must be available in each generation" is his statement of the first half; the second half is that selection pressure, acting continuously, holds a population near a current optimum and prevents most of that variation from establishing. Nietzsche made the same observation from the breeder's side, that species given superabundant nourishment "tend in the most marked way to develop variation" [2]. Read that way, an abundance of resources is less a cause of diversification than a withdrawal of what had been suppressing it, and the *speciation rate* becomes a function of how much room is left.

That is a claim about the fossil record as much as about breeding. The signature — high early diversification after the opening of ecological space, then a slowing — is the one named by punctuated equilibria [3], and it is the one density-dependent diversification models are built to produce [4]. Whether it is the typical pattern, and at what taxonomic level, is not settled here.

1.2.2 The push of the past

There is an existing account of why diversification rates appear high at the origin of a clade and decline towards the present, and the model below is a response to it rather than a competitor. Mann and Budd, reviewing the phenomenon introduced by Nee and co-workers [5], name it the push of the past [6]:

The POTPa emerges as a feature of diversification by the fact that all modern clades (tautologically) survived until the present day. This singles them out from the total population of clades that could be generated from any particular pair of background birth and death rates: clades that happened by chance to start off with higher net rates of diversification have better-than-average chances of surviving until the present day. As Nee et al. (1994a) put it, such clades "got off to a flying start"; and they accumulate species faster than one would expect. Once clades become established, they are less vulnerable to random changes in the observed diversification rate, and this value therefore tends to revert to the background rate through time. Long-lived clades thus tend to show high observed rates of diversification at their origin, which then decrease to their long-term average as the present is approached.

The push of the past is a survivorship effect: the rates never move, and the apparent decline is an artefact of conditioning on survival to the present. The model of this section supposes instead that the rate really does move, declining because the ecological room that permitted the early diversification is used up. The two accounts are not exclusive and nothing here argues against the first. What the model offers is a second mechanism with the same signature, and a distribution against which it might one day be told apart from the first [7].

1.2.3 The model

Let $N(t)$ be the total abundance of individuals across all species, taken as a real number on the grounds that it is large compared with the number of species, and let it grow logistically,

$$\frac{dN}{dt} = \varrho N \left(1 - \frac{N}{K}\right), \quad N(0) = N_0, \quad 0 < N_0 < K, \quad (1.1)$$

with K the carrying capacity of the environment and ϱ the growth rate. Its solution is

$$N(t) = \frac{K e^{\varrho t}}{\xi + e^{\varrho t}}, \quad \xi = \frac{K}{N_0} - 1. \quad (1.2)$$

Let S_t count species, with $S_0 = 1$. Each existing species gives rise to a new one independently at the per-species rate $\beta(t)$, and no species becomes extinct, so the transitions in a vanishing interval are

$$\begin{aligned} \mathbb{P}(S_{t+\Delta t} - S_t = 1 \mid S_t = n) &= \beta(t) n \Delta t + o(\Delta t), \\ \mathbb{P}(S_{t+\Delta t} - S_t = 0 \mid S_t = n) &= 1 - \beta(t) n \Delta t + o(\Delta t). \end{aligned} \quad (1.3)$$

This is the Yule process of ?? with its constant rate replaced by a function of time; setting $\beta(t) = \beta$ recovers it exactly. The substitution is not new in itself — taking $\beta(t) = e^{-t}$ makes the mean of S_t follow Gompertz’s 1825 law [8] — and the Yule process was itself introduced in 1925 to model exactly this quantity, the number of species within a genus [9]. What is being tried here is a rate tied to the logistic environment:

$$\beta(t) = \sigma(K - N(t)), \quad (1.4)$$

which by (1.2) is

$$\beta(t) = \sigma K \frac{\xi}{\xi + e^{\varrho t}}, \quad (1.5)$$

decreasing from $\sigma(K - N_0)$ to zero as the environment fills. The coupling runs one way only: the environment sets the speciation clock, and S_t does not act back on $N(t)$. The same model appears as a worked example in ??, where its mean is recorded; what is added here is its distribution.

Remark 1.1 (What the rate law is, and is not, claiming). (1.4) treats resource availability as an *index* of selective pressure and not as a mechanism. It says that when there is room, variants persist more often, and it says nothing about how. The model is correspondingly agnostic about the direction of causation: it does not distinguish logistic expansion driving diversification from logistic expansion merely creating the space in which diversification happens.

Underneath that sits a question of timescale. If population expansion is orders of magnitude faster than speciation — which for most plausible parameter values it is — then the logistic phase is over before the species count has moved appreciably, and (1.4) is better read as setting the initial condition for a density-dependent diversification model than as driving the rate through time [4]. Nothing below depends on resolving that, since the mathematics of [proposition 1.2](#) holds for any locally integrable β , and the question is which β the biology warrants.

1.2.4 The marginal law

The first result is that the rate law enters the distribution of S_t only through its integral, and that the resulting law is geometric. Write

$$\mathcal{B}(t) = \int_0^t \beta(u) du \quad (1.6)$$

for the integrated speciation rate, and assume only that β is locally integrable and non-negative.

Proposition 1.2 (Marginal law of an inhomogeneous Yule process). *Let S_t be a pure birth process on $\{1, 2, \dots\}$ with $S_0 = 1$ and time-dependent per-capita rate $\beta(t)$. Then S_t is geometric on $\{1, 2, \dots\}$ with parameter $e^{-\mathcal{B}(t)}$:*

$$\mathbb{P}(S_t = n) = e^{-\mathcal{B}(t)} (1 - e^{-\mathcal{B}(t)})^{n-1}, \quad n \geq 1, \quad (1.7)$$

with

$$\mathbb{E}(S_t) = e^{\mathcal{B}(t)}, \quad \text{Var}(S_t) = e^{\mathcal{B}(t)} (e^{\mathcal{B}(t)} - 1).$$

The proof is the generating-function computation of [section 1.C](#): the forward Kolmogorov equation still yields to the method of characteristics, and the probability generating function is that of a geometric law on $\{1, 2, \dots\}$ with success parameter $e^{-\mathcal{B}(t)}$. Three things are worth recording here. The first is that the constant-rate Yule law ?? is the case $\mathcal{B}(t) = \beta t$, so nothing is being rebuilt: the substitution $\beta t \mapsto \mathcal{B}(t)$ is exact and not merely a convenience. The second is that the argument nowhere uses the form of β , only that its integral exists, so two rate laws with the same integral to time t give the same distribution at t however differently they arrived there. The third is that the variance falls out of the same line, and had not been recorded before. The standard deviation is $e^{\mathcal{B}} \sqrt{1 - e^{-\mathcal{B}}}$, of the same order as the mean, so to fit this model to a single lineage is to fit a mean to one draw from a law whose coefficient of variation is close to one.

1.2.5 The mean, directly

With the proposition in hand $\mathbb{E}(S_t) = e^{\mathcal{B}(t)}$ is a corollary, and it is also ?? with $\mu \equiv 0$. The direct derivation is retained as a check on the integral that follows.

Over a vanishing interval the mean after is the mean before plus the expected change, and by (1.3) the expected change is $\beta(t)\mathbb{E}(S_t) \Delta t + o(\Delta t)$. Dividing by Δt and letting it vanish,

$$\frac{d}{dt} \mathbb{E}(S_t) = \beta(t) \mathbb{E}(S_t), \quad (1.8)$$

which separates. With $\mathbb{E}(S_0) = 1$ the solution is $\mathbb{E}(S_t) = e^{\mathcal{B}(t)}$, and for the logistic rate law (1.5) the integral is elementary:

$$\int \frac{dt}{\xi + e^{\varrho t}} = \frac{1}{\xi} \left(t - \frac{1}{\varrho} \ln(\xi + e^{\varrho t}) \right), \quad (1.9)$$

so that

$$\mathcal{B}(t) = \sigma K \left(t - \frac{1}{\varrho} \ln(\xi + e^{\varrho t}) + \frac{1}{\varrho} \ln(\xi + 1) \right), \quad \mathbb{E}(S_t) = e^{\sigma K t} \left(\frac{\xi + 1}{\xi + e^{\varrho t}} \right)^{\sigma K / \varrho}. \quad (1.10)$$

1.2.6 The limit law

The question the model was built to answer is what happens at late time, when β has fallen to nothing and the species count has stopped moving. Since S_t is non-decreasing and its mean is bounded, it converges almost surely to a finite limit; the content of the following is that the limit is proper and its law is known.

Corollary 1.3 (Limit law under the logistic rate). *Under (1.4), $\mathcal{B}(t)$ converges as $t \rightarrow \infty$, to*

$$\mathcal{B}(\infty) = \frac{\sigma K}{\varrho} \ln \left(\frac{K}{N_0} \right), \quad (1.11)$$

and S_∞ is geometric on $\{1, 2, \dots\}$ with

$$\mathbb{E}(S_\infty) = \left(\frac{K}{N_0} \right)^{\sigma K / \varrho}. \quad (1.12)$$

Proof. From (1.10),

$$t - \frac{1}{\varrho} \ln(\xi + e^{\varrho t}) = -\frac{1}{\varrho} \ln(1 + \xi e^{-\varrho t}) \longrightarrow 0,$$

so $\mathcal{B}(t) \rightarrow (\sigma K / \varrho) \ln(\xi + 1)$, and $\xi + 1 = K / N_0$. The law is (1.7) with $\mathcal{B}(t)$ replaced by its limit, which is legitimate because S_t increases to S_∞ and the geometric probabilities are continuous in the parameter. $\square$

The dependence on the parameters is now explicit, and it is simpler than the five constants $\sigma, K, \varrho, N_0, \xi$ suggest: only two dimensionless groups survive. The first is K/N_0 , the room the genus had at the outset, so that a clade founded into a nearly empty environment integrates more unused capacity than one founded into a full one and ends with more species. The second is $\sigma K/\varrho$, the speciation capacity per unit of population growth rate, which sets how much of that room is converted before it closes; doubling the growth rate and the speciation coefficient together changes nothing. That is the answer to the question posed but not settled when the model was written — how the asymptotic behaviour depends on the parameters and their interplay — and it is an answer about the whole distribution rather than about the mean alone.

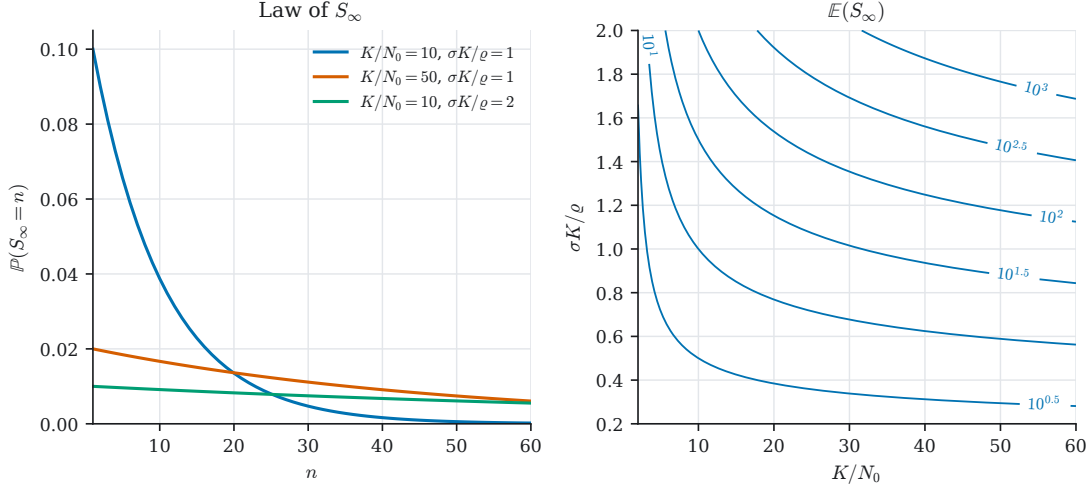


Figure 1.2: What [corollary 1.3](#) says, drawn from the closed form. Panel (a): the law of S_∞ is geometric, so most of its mass sits at small counts however large the mean, and its coefficient of variation is close to one — fitting this model to a single clade is fitting a mean to one draw from that. Panel (b): the expected final count $E(S_\infty) = (K/N_0)^{\sigma K/\varrho}$ depends on the five constants $\sigma, K, \varrho, N_0, \xi$ only through the two dimensionless groups on the axes, so every point on a contour is a genus with the same expected diversity reached from different parameters.

1.2.7 The other rate laws

The form (1.4) was one of five candidates written down when the model was set up. The others were

$$\begin{aligned} \beta_2(t) &= \sigma(K - N(t)) + \eta, & \beta_3(t) &= \sigma \frac{dN}{dt}, \\ \beta_4(t) &= \sigma \frac{dN}{dt} + \eta, & \beta_5(t) &= \sigma(K_t - N(t)), \end{aligned} \quad (1.13)$$

with η a constant floor and K_t a capacity switching at random. The motivation for β_3 and β_4 is that a mutation event needs a replication event, so speciation might track the rate of population increase rather than the room remaining; the floor η in β_2 and β_4 is a baseline propensity to vary that survives the environment filling up. [Proposition 1.2](#) settles two of the four immediately, because it asks only for an integral.

For β_3 the integral is the total change in abundance:

$$\mathcal{B}(\infty) = \sigma \int_0^\infty N'(u) du = \sigma(K - N_0), \quad (1.14)$$

so S_∞ is again geometric, with mean $e^{\sigma(K - N_0)}$. The comparison with (1.12) is worth a moment. Under β_1 the limiting mean is a power of K/N_0 and under β_3 an exponential of $K - N_0$, so the

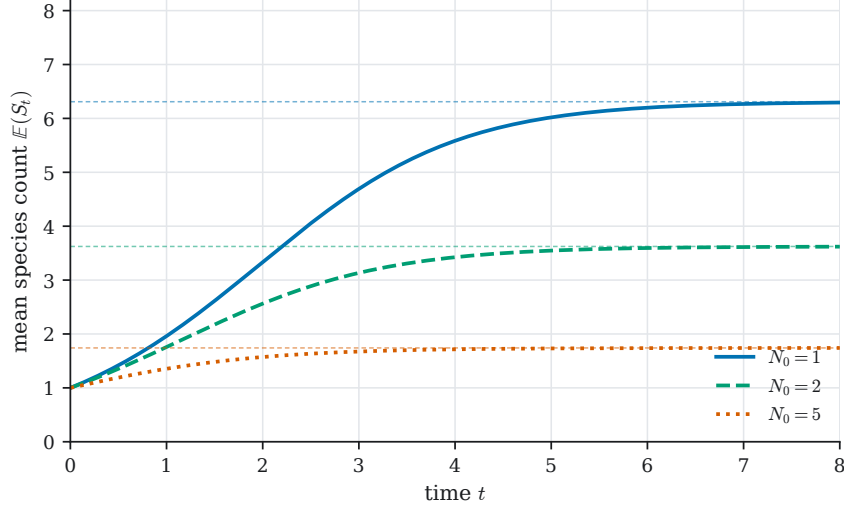


Figure 1.3: Mean species count (1.10) for $N_0 = 1, 2, 5$, with $K = 10$, $\varrho = 1$ and $\sigma = 0.08$; dashed lines are the limits $(K/N_0)^{\sigma K/\varrho}$. A smaller founding abundance leaves more integrated unused capacity, so the limiting mean is higher. This is the pure-birth model of (1.3), not the birth–death numerical variant recorded at ??.

first is sensitive to the *ratio* of capacity to founding abundance and the second to their *difference*; a model fitted under one and read under the other would be badly wrong for large K .

For β_2 and β_4 the floor makes $\mathcal{B}(t) \sim \eta t$, which diverges, and there is then no limit law: S_t remains geometric at every finite time with parameter $e^{-\mathcal{B}(t)} \rightarrow 0$, and the species count grows without bound, eventually at the constant-rate Yule rate η . That is a statement about the model, not a defect of the analysis: a floor on the speciation rate says diversification never stops, and a process that never stops has no final species count for a distribution to be about.

β_5 is the one that remains genuinely open, and the proposition says why. If K_t switches at random then $\mathcal{B}(t)$ is itself a random variable, and (1.7) holds conditionally on it. Unconditionally S_t is a *mixture* of geometric laws over the law of $\mathcal{B}(t)$,

$$\mathbb{P}(S_t = n) = \mathbb{E}\left(e^{-\mathcal{B}(t)}(1 - e^{-\mathcal{B}(t)})^{n-1}\right), \quad (1.15)$$

so that the problem becomes finding the law of the integrated rate under the switching regime. For a finite-state Markov switching K_t that is a Markov-modulated integral and is tractable; for anything else it is not obviously so. Of five candidate rate laws, two are now closed, two are explained non-results, and one is a well-posed problem.

The pattern across all five is the same one: nothing about a moving speciation rate is difficult so long as its integral is deterministic, and everything about it is difficult once the integral is not.

Open questions.

1. **The mixture at β_5 .** What is the law of $\mathcal{B}(t)$ when K_t switches between finitely many values at exponential rates, and what does the resulting mixture (1.15) do to the tail of S_t ? This is the question in this section most worth taking up, and it asks for a Markov-modulated integral rather than for a new model.
2. **Identifiability.** Since only $\mathcal{B}(t)$ enters (1.7), no observation of S_t at a single time can separate σ , K and ϱ beyond the two groups of (1.12). What a time series of S_t at several times does identify, and whether β_1 and β_3 can be told apart from one, is not known.
3. **The timescale question of remark 1.1.** Whether population expansion and speciation

are separated by orders of magnitude in any clade for which both are measurable, and therefore whether the model describes a rate or an initial condition, is an empirical question this chapter cannot settle.

1.3 Competitive reversal: the Pimentel model

The speciation model of the previous section lets variation out when there is room. This one asks the complement: what happens when two populations share the room, and the minority is the one under selection.

1.3.1 The experiment

Pimentel and colleagues devised and ran an experiment in which two populations of flies are pitted against one another in a fight for domination and, ultimately, survival [10, 11]. Place a colony of houseflies (*Musca domestica*) in one side of a boxed enclosure and a competing number of blowflies (*Phaenicia*) at the other, and give them an ample but finite supply of food. Before long one of the species dominates the niche. The arrangement is unstable, and either the housefly or the blowfly population eventually sees off the other.

Pimentel and his colleagues did something clever to augment the dynamics. They contrived an enclosure — the original box, but with a myriad of internal sub-compartments, a three-dimensional maze of tunnels and small chambers. That arrangement gives the enclosure a desirable property: with the populations initially placed at opposing sides, communication by contact is delayed by a period greater than the average lifespan of either species. In the months taken for a species to spread from one side to the other, multiple generations ensue.

Why is that desirable? By slowing the potential spread, Pimentel imposes the constraint that a total domination, if one is arrived at, cannot be won in one or a few fly lifespans. The war must be won over succeeding generations, grandchildren’s grandchildren all fighting for the same cause, and that is what allows the hand of Darwinian selection to come into play.

The observed dynamic was the following. Blowflies were pushed almost to extinction by the initially superior numbers of *Musca domestica*, and then the tide turned: from the brink, the blowfly underdogs clawed back their footing and eventually overthrew the advances of their adversaries. The hypothesis is that during this squeeze of reduced population the blowflies began to select, among their limited numbers, for the abilities favouring direct competition with their adversaries. The houseflies, doing well as they were, were subject to no such channelling selection; they became complacent while their foes grew strong, cut into shape by a harshly oppressive environment.

It has not been achieved experimentally, but Pimentel proposed that with the enclosure arranged correctly an oscillation could be set up, in which the upper hand passes continually from one side to the other, each being pushed to the brink and moulded to competitive advantage in times of scarcity — a continual arms race for survival. The question this section asks is whether a Markov chain with evolving blocking probabilities can reverse in the same way.

1.3.2 The model

Consider two populations competing for space in a finite domain they both occupy. Let A_t count one of them, labelled a , and B_t the other, labelled b , both in discrete time $t \in \mathbb{N} \cup \{0\}$. Finiteness of the enclosure is imposed by $A_t + B_t \leq N$ for a maximum capacity N , and with the initial condition $A_0 = B_0 = N/2$ for even N the two counts are tied by $B_t = N - A_t$ throughout,

so it is enough to follow A_t . Its transition probabilities are

$$\mathbb{P}(A_{t+1} = j \mid A_t = i) = \begin{cases} q_a = \frac{i}{N}(1 - p_b), & j = i + 1, \\ q_b = \left(1 - \frac{i}{N}\right)(1 - p_a), & j = i - 1, \\ q_c = \frac{i}{N}p_b + \left(1 - \frac{i}{N}\right)p_a, & j = i, \\ 0 & \text{otherwise,} \end{cases} \quad (1.16)$$

for $0 < i < N$, with $i = 0$ and $i = N$ absorbing.

The three cases come from the following step rule, viewed from the perspective of population a .

Whose move. With probability i/N population a attempts to advance; with probability $1 - i/N$ the opponent attempts to, potentially forcing a back one.

Advancing. If a has decided to advance it does so, unless it is blocked, which happens with probability p_b .

Being advanced upon. If the opponent is advancing, a blocks with probability p_a ; if it fails to, it is forced back a step.

Stalemate. If a block is effective, both parties remain where they are, and $A_{t+1} = A_t$.

The capacity to block is the model's evolving quantity. Let it depend on a *resistance* R_k for $k = a, b$ through

$$p_a = \max \left\{ 0, 1 - \frac{R_b}{R_a} \right\}, \quad p_b = \max \left\{ 0, 1 - \frac{R_a}{R_b} \right\}, \quad (1.17)$$

so that at most one of the two is non-zero: whichever population has the higher resistance blocks, and the other does not. If $R_a \gg R_b$ then p_a is close to one and p_b is zero.

Resistances are driven up when the associated population is in a minority. They are updated at each step by

$$R_{k,t+1} = R_{k,t} + \varepsilon \Delta R_{k,t}, \quad (1.18)$$

at an evolution rate ε , with

$$\Delta R_{a,t} = \max \left\{ 0, \left(\frac{N - A_t}{A_t} \right)^\alpha \right\}, \quad \Delta R_{b,t} = \max \left\{ 0, \left(\frac{A_t}{N - A_t} \right)^\alpha \right\}. \quad (1.19)$$

Each increment grows as its population is squeezed towards its absorbing boundary, and is larger for the population in the minority. Only the ratio R_b/R_a enters (1.17), so what drives the blocking probabilities is the *difference* between the two increments rather than either alone. The mechanism intended is the one Pimentel proposed: most of the competitive interactions experienced by members of the smaller population are with the opposition, and the drive for selection towards competitive ability is correspondingly stronger, while the majority population meets mostly its own kind and is under no such pressure. The exponent α sharpens that drive near the boundaries. Taking $\alpha = 1$ makes the increment simply proportional to the ratio of interspecific to intraspecific encounters, which is the physical case; $\alpha > 1$ is retained because it is what makes the criticality of [section 1.3.4](#) visible.

The initial resistances need saying, because the model is not scale-free in them. Since (1.17) depends on R_a and R_b only through their ratio, while (1.18) adds an absolute increment, scaling $R_{a,0}$, $R_{b,0}$ and ε by a common factor leaves every trajectory unchanged: the two enter only through ε/R_0 . Taking $R_{a,0} = R_{b,0} = 1$ throughout therefore costs nothing and fixes what ε means.

Two things about (1.16), (1.17) and (1.19) are worth naming, since they are what make this a chapter about moving rates. The first is that the step probabilities are functions of the resistances and the resistances are functions of the state, so the transition kernel at time t depends on the whole path up to t and not on A_t alone. The second follows: the triple $(A_t, R_{a,t}, R_{b,t})$ is Markov, and A_t by itself is not.

1.3.3 Results

Simulated at $\alpha = 1$, the model does what Pimentel observed. Both realisations in figure 1.4 take the same shape: one population reaches a position of real advantage, and does not keep it. Trained by interspecific competition, the oppressed population makes a striking comeback and overthrows its adversary completely. At this exponent the reversal happens once or twice and then the chain is absorbed, so what Pimentel hoped for arrives as an episode rather than as a cycle.

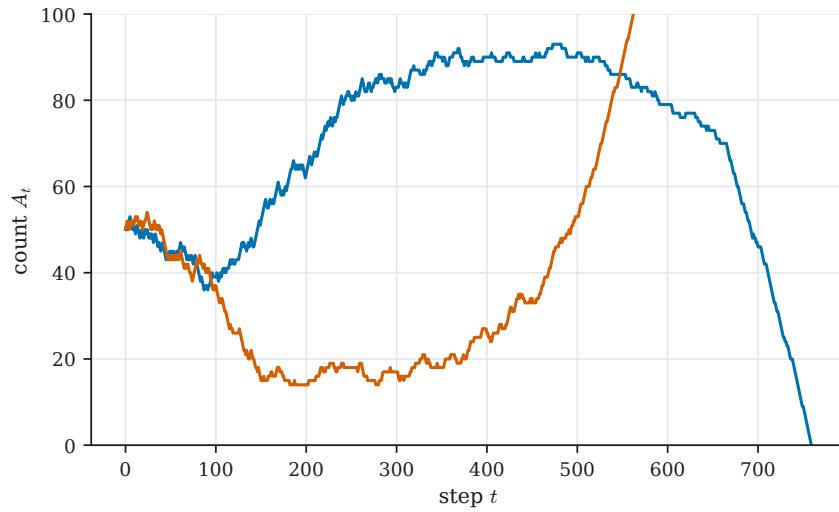


Figure 1.4: Two realisations of the Pimentel chain at $\alpha = 1$, $N = 100$, $\varepsilon = 10$, the physical case. Each reverses: an early advantage is not kept, and the two histories, started from the same symmetric state, absorb at opposite boundaries.

Raising α sharpens the resistance drive near the boundaries, and with it the sensitivity of a trajectory to where it starts. Figure 1.5 makes that comparison at $\alpha = 10$ and $N = 10^5$, running two independent realisations from each of two initial conditions: one with the populations evenly matched, one with a bias imposed at the outset. Started well matched, the two realisations diverge substantially; started biased, they stay close for far longer. What differs is not the noise, which is the same in both, but where the trajectory sits relative to the state at which the chain has no preferred direction.

Raising α also buys far more of the behaviour the model was built for. Where $\alpha = 1$ gives one reversal or two, $\alpha = 10$ buys of order ten reversals before fixation, and the swing grows rather than decays. Figure 1.5 under-shows that run: the full excursion, at the same parameters, continues for several million further steps before absorption. The oscillation Pimentel conjectured is therefore not absent from the model, only impermanent. Every run here still fixates in the end, and what raising α moves is how long the alternation lasts before it does.

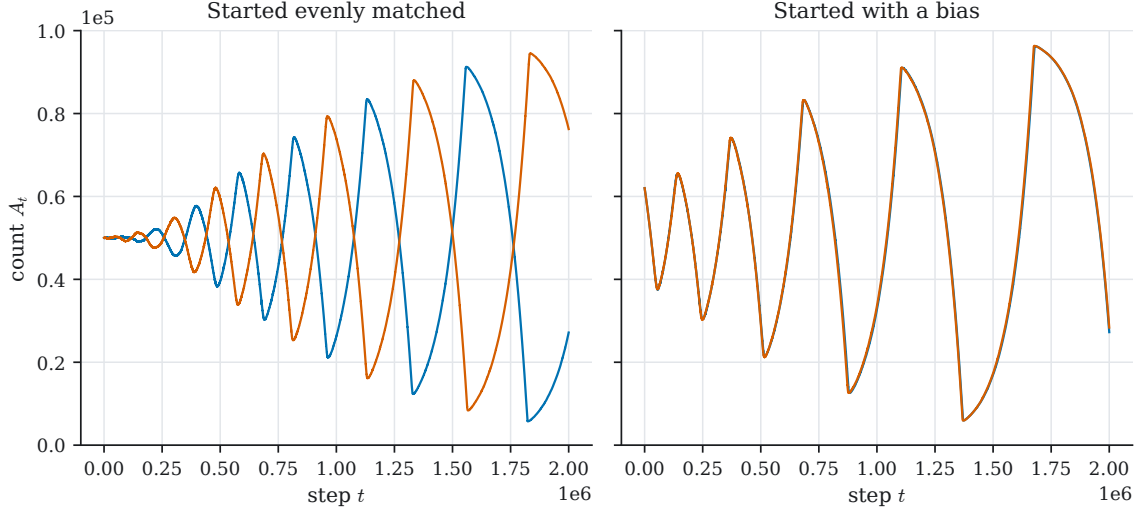


Figure 1.5: Two independent realisations from each of two initial conditions, at $\alpha = 10$, $N = 10^5$, $\varepsilon = 1$, over 2×10^6 steps. Panel (a): started evenly matched, and so near the critical state, the runs separate early and stay apart. Panel (b): started with one in the majority, and so away from it, they track one another. Proximity to the critical state, not the size of the noise, decides how far apart two histories of the same model end up.

1.3.4 The moving critical point

That state can be located exactly. The chain is at its *critical point* when a step up and a step down are equally likely, that is when $q_a = q_b$ in (1.16). Writing i_c for the state at which this holds and using the mutual exclusivity of p_a and p_b ,

$$i_c = \begin{cases} N \frac{1 - p_a}{2 - p_a}, & p_a > 0 \text{ and } p_b = 0, \\ N \frac{1}{2 - p_b}, & p_b > 0 \text{ and } p_a = 0. \end{cases} \quad (1.20)$$

When neither population blocks, $p_a = p_b = 0$ and $i_c = N/2$, the symmetric point. Blocking moves it off centre, and moves it towards the boundary the *leading* population is heading for: if a is in the majority then it is b that accumulates resistance, so $p_b > 0$ and $i_c = N/(2 - p_b)$ climbs towards N . And i_c is an *unstable* point — comparing q_a with q_b in (1.16) gives $q_a > q_b$ exactly when $i > i_c$, in both cases of (1.20) — so the chain is carried away from it rather than returned to it. A reversal is what happens when i_c , climbing after the leader, overtakes the trajectory, and the drift that was carrying the majority forward begins to carry it back.

(1.20) is less analytic than it looks. Since p_a and p_b depend on R_a and R_b , and those are accumulated along a stochastic trajectory by (1.18), i_c is a function of the path and not of the parameters alone. What it is not is intractable. Given a realisation, the sequence $i_{c,t}$ is computed step by step at the cost of the simulation itself and can then be drawn over the trajectory that produced it, which is what figure 1.6 does.

Read against figure 1.5, figure 1.6 makes the mechanism plain: trajectories diverge exactly where A_t is near $i_{c,t}$, because there the drift that would otherwise restore them is absent. That is what the resistance dynamics buy. An ordinary random walk on $\{0, \dots, N\}$ is critical only if its step probabilities happen to be balanced, whereas here the resistance update *drags the critical point after the leading population* and so keeps returning the process to a state at which its own future is undetermined. The reversal Pimentel observed is therefore not a deterministic consequence of the squeeze; it is what a squeeze does to a chain that the squeeze itself has made critical.

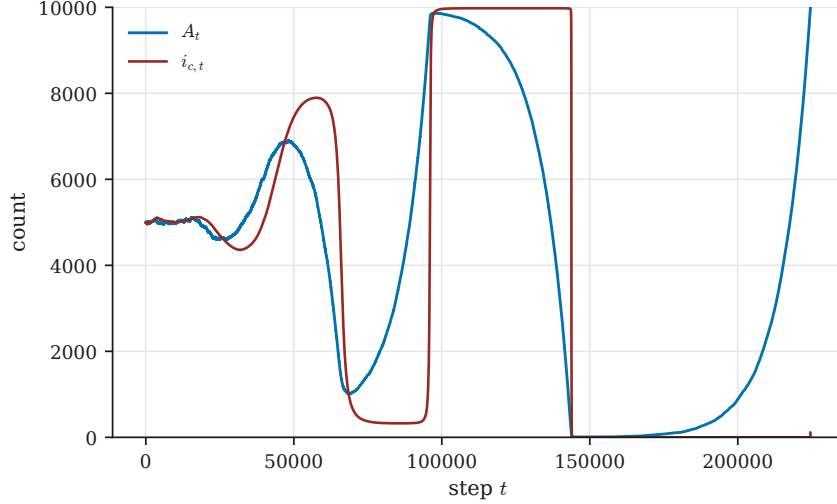


Figure 1.6: A single realisation at $\alpha = 3$, $N = 10^4$, $\varepsilon = 1$, with the critical state $i_{c,t}$ of (1.20) overlaid. The critical state does not stay put. It is dragged towards whichever boundary the leading population is heading for, and saturates near 0 or near N while that population is well ahead; each reversal of A_t follows $i_{c,t}$ crossing it.

1.3.5 A spatial version

The model above has no geometry, where Pimentel's enclosure had a great deal of it. A spatial version was therefore written as a cellular automaton, in C++ using the SFML graphics library, each cell holding one of the two populations and resistances being inherited and updated locally by the same rule; figure 1.7 shows six frames from it. A model of evolutionary reversal in competitive dominance had been proposed before [12]; the model here was not derived from it and differs in form.

1.3.6 Coupling speciation to competition

Section 1.3 makes a losing population strong, the squeeze being what drives its resistance up. This subsection asks the complement, whether a winning population can be made weak by a mechanism of the same kind; the hypothesis it states is that excess variation weakens an established type, so that a population which is winning diversifies and thereby loses. The claim is Nietzsche's, put into a model: a type is made hard by long struggle against constant unfavourable conditions, favourable conditions relax the constraint which made it hard, and the resulting exuberance of variation is itself the mechanism of decline [2]. What is worth retaining is the shape of the claim: that the decline is *endogenous*, a consequence of the success preceding it rather than a response to any external shock.

The model is the kernel of section 1.3.2, recast so that the quantity under selection is fitness rather than resistance, with a species count carried alongside and allowed to act back on that fitness. Written as step probabilities for A_t ,

$$\mathbb{P}(\Delta A_t = 1) = \frac{A_t}{N} \frac{f_a}{\bar{f}}, \quad \mathbb{P}(\Delta A_t = -1) = \left(1 - \frac{A_t}{N}\right) \frac{f_b}{\bar{f}}, \quad (1.21)$$

with $\bar{f}$ the mean fitness across the domain and, under the Moran assumption, $A_t = N - B_t$. Each probability is again the chance of meeting a member of the opposing group multiplied by the chance of prevailing against them, as in (1.16); what has changed is that prevailing is now governed by a fitness ratio rather than by a blocking probability. Fitness is driven up by

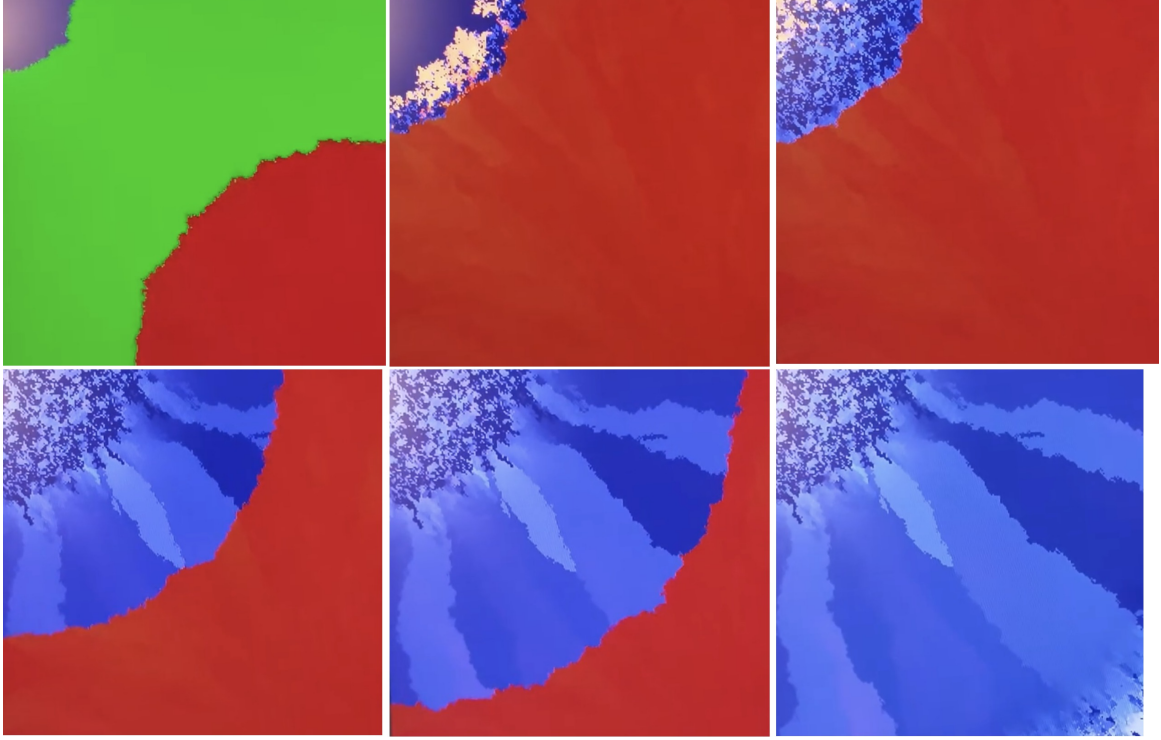


Figure 1.7: Six frames from the spatial version, left to right along the top row then the bottom: advance, squeeze, fringe, crossing, recovery, fixation. Unoccupied cells are green in the first frame. Red advances until blue is confined to one corner, where a pale fringe of elevated resistance appears; blue then crosses back and takes the domain. The reversal is spatial as well as numerical.

minority status exactly as resistance was:

$$f_a(t + \Delta t) = f_a(t) + \varepsilon \left(\frac{B_t}{A_t} \right)^\alpha, \quad f_b(t + \Delta t) = f_b(t) + \varepsilon \left(\frac{A_t}{B_t} \right)^\alpha. \quad (1.22)$$

Fitness here is the same evolving quantity as the resistance R_a, R_b of the previous section, under a second name: the update rule and its exponent are identical, and the increment coefficient is ε in both. The two kernels are not literally identical, since (1.16) has a stalemate outcome and (1.21) has none; both are carried because the fitness form is the one that admits the coupling below, in which a scalar can be added to and subtracted from.

Now give each population its own species count, $S_t^{(a)}$ and $S_t^{(b)}$, evolving as birth-death processes with constant per-species extinction μ_a and a speciation rate that rises with both dominance and fitness:

$$\mathbb{P}(\Delta S_t^{(a)} = 1) = \beta_a S_t^{(a)} \Delta t, \quad \mathbb{P}(\Delta S_t^{(a)} = -1) = \mu_a S_t^{(a)} \Delta t, \quad \beta_a = \left(\frac{A_t}{N} \right)^\zeta f_a, \quad (1.23)$$

and correspondingly for b with A_t replaced by B_t , the exponent ζ controlling how firmly dominance must be established before variation is released. This is the premise of section 1.2 transplanted into a competitive setting: variation is in constant supply, and what releases it is the easing of whatever was suppressing it, measured here by the share of the domain a population controls.

The coupling that closes the loop is a penalty on fitness proportional to the species count:

$$f_a(t + \Delta t) = f_a(t) + \varepsilon \left(\frac{B_t}{A_t} \right)^\alpha - \gamma S_t^{(a)}, \quad f_b(t + \Delta t) = f_b(t) + \varepsilon \left(\frac{A_t}{B_t} \right)^\alpha - \gamma S_t^{(b)}, \quad (1.24)$$

with γ the amount by which the inefficiency of carrying excess variation weighs on competitive performance.

That is the whole specification, and it has never been simulated. Both routes to reversal are present in it at once: the squeeze that strengthens the loser, through (1.22), and the variation that weakens the winner, through the penalty of (1.24). Both favour whichever population is behind, so the model does not say in advance which of them does the work, and that is what makes it worth running.

Open questions.

1. **Whether collapse by variation happens at all.** Does the $-\gamma S_t^{(a)}$ term in (1.24) ever produce a reversal that the squeeze-driven mechanism of section 1.3 would not have produced unaided, or does it only slow that one down? Separating the two means holding everything else fixed and comparing runs at $\gamma = 0$ against runs at $\gamma > 0$, which asks for simulation and for nothing else.
2. **The relevant regime.** Collapse by variation should require $\gamma S_t^{(a)}$ to grow comparable with the accumulated fitness advantage over the timescale on which $S_t^{(a)}$ itself grows, which ties γ , ζ and μ_a together. Whether that regime is a thin sliver of parameter space or a broad band of it follows directly from the first question.

Open questions.

1. **Fixation.** The fixation probability and the expected time to fixation, as functions of (α, ε, N) , were never computed. The first-step machinery of ?? applies to the enlarged chain $(A_t, R_{a,t}, R_{b,t})$, which is where the work would start.
2. **Whether the oscillation persists.** Every realisation run here eventually fixated, at $\alpha = 1$ after an episode and at $\alpha = 10$ after of order ten reversals. So the question is not whether the alternation exists but whether it can be made to last: does the expected number of reversals before fixation grow without bound in α , ε or N , or does it saturate? This is the sharpest question in the section, because it is the one the model was built to answer.
3. **A diffusion limit.** For large N there should be a diffusion approximation in which (1.20) becomes a moving unstable equilibrium and the divergence of nearby trajectories is a statement about a local Lyapunov exponent. That would turn the qualitative observation of figure 1.6 into a rate.

1.4 Rise, run, ruin, rejuvenation

1.4.1 The cycle the model is for

Where the two competition models work inside an arena of fixed capacity, this one replaces the capacity with a flow, and the flow buys a cycle in four movements. *Potential* accumulates while it goes unused; replicative agencies proliferate into the surplus once there is one; their proliferation consumes the potential until scarcity brings a catastrophe in which the agencies are lost; and the potential rebuilds until a revival restarts the wave.

The idea is Prigogine's. In a system through which energy or matter flows, structures can emerge that appear to run against the second law: order appears locally, as a spontaneous reduction of entropy, sustained by the throughput and destroyed if the throughput stops [13]. Such *dissipative structures* are characterised by chaotic dynamics, in which small fluctuations are amplified into the details of the eventual form. Whether the fragmentation of a genus into species is an instance of one has been argued [14, 15], and the model below does not settle it;

the claim made here is the weaker one, that the four-part cycle is a recognisable shape worth writing down as a Markov chain. The pair $(X_t, V(t))$ is a birth–death–catastrophe process with catastrophe rate $\delta X_t/V(t)$ and an auxiliary ODE for V , rather than a new class of object.

1.4.2 The model

Let X_t count replicative agencies and $V(t)$ the accumulated potential. Potential flows in at a constant rate ϕ and is consumed at a rate c per agency:

$$\frac{dV}{dt} = \phi - c X_t. \quad (1.25)$$

Given V , the count X_t moves by four channels, of which the second is the one that carries the model:

$$\begin{aligned} \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda X_t \Delta t + o(\Delta t) && \text{proliferation,} \\ \mathbb{P}(X_{t+\Delta t} = 0 \mid X_t) &= \delta \frac{X_t}{V(t)} \Delta t + o(\Delta t) && \text{catastrophe,} \\ \mathbb{P}(X_{t+\Delta t} = 1 \mid X_t = 0) &= \omega \Delta t + o(\Delta t) && \text{revival,} \\ \mathbb{P}(X_{t+\Delta t} = X_t) &= 1 - \left(\lambda X_t + \delta \frac{X_t}{V(t)} + \omega \mathbf{1}\{X_t = 0\} \right) \Delta t + o(\Delta t) && \text{no event.} \end{aligned} \quad (1.26)$$

The catastrophe channel is the model’s whole content. Its rate is not δX_t , as it would be in the birth–death–catastrophe process of the chapter on the birth–death–catastrophe process, but $\delta X_t/V(t)$, so that catastrophe is driven by per-capita scarcity: a large population sitting on a large potential is no more at risk than a small one on a small potential, while a population that has consumed its potential is at risk in proportion to how thoroughly it has done so. That is what makes the cycle close rather than run once. Revival at rate ω out of the empty state keeps 0 from being absorbing, and is what makes the process recurrent.

The potential cannot in fact be exhausted, which is what makes the model well posed. As $V \downarrow 0$ with $X_t \geq 1$ the catastrophe hazard $\delta X_t/V(t)$ behaves like $\delta/(c(t^* - t))$, where t^* is the time (1.25) would reach zero; that integrates to infinity, so catastrophe arrives almost surely first and V stays positive for all time.

Neither nonlinearity here is available to the machinery of the preceding chapters. The catastrophe rate depends on V , and V depends on the whole history of X through (1.25), so that the pair $(X_t, V(t))$ is Markov while X_t alone is not and no generating function for X_t closes.

1.4.3 What the simulations show

Figures 1.8 and 1.9 are exact realisations. They show the cycle running, and they show that its period is not a constant.

What X_t counts is deliberately left open. It may be individuals in a population, species within a genus, or autonomous units within an organisation, and the model does not distinguish them: it is a proxy for order, or for locally reduced entropy, whose units are whatever the application supplies. That generality is also the model’s principal weakness, and section 1.8.2 returns to it.

Open questions.

1. **The cycle period.** The distribution of the interval between successive catastrophes is a first-passage problem for the pair $(X_t, V(t))$, and is not known even in its mean. The figures establish that the period has substantial spread; how much, and how it depends on ϕ , c , λ and δ , is the quantity to go after.

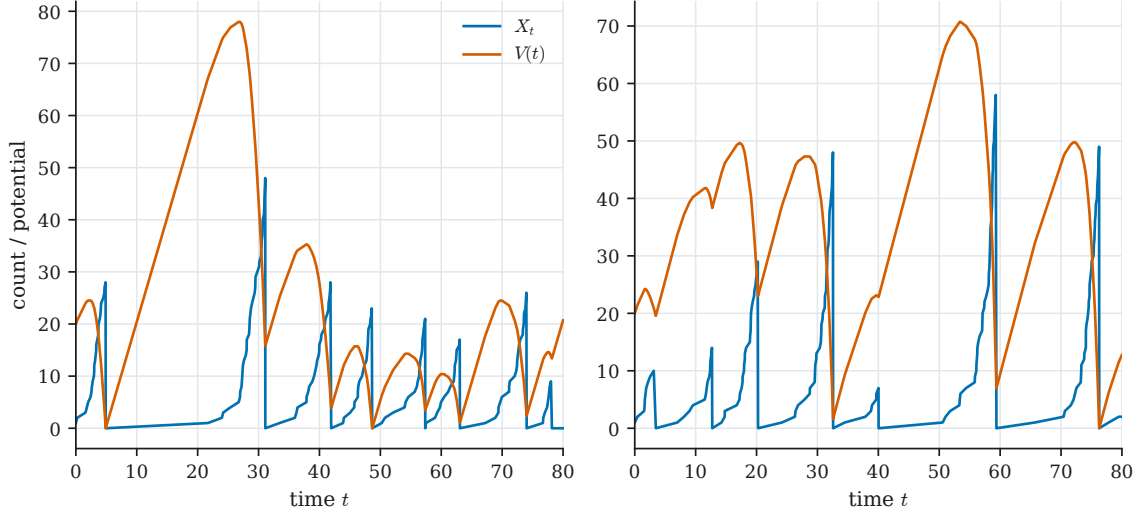


Figure 1.8: Two realisations of (1.25) and (1.26): potential $V(t)$ in vermillion, agency count X_t in blue ($\phi = 4$, $c = 0.8$, $\lambda = 0.5$, $\delta = 0.25$, $\omega = 0.15$). Every cycle has the same four parts in the same order — potential builds while X_t is small, X_t grows into the surplus, the surplus is consumed, the collapse follows — and no two are copies of one another.

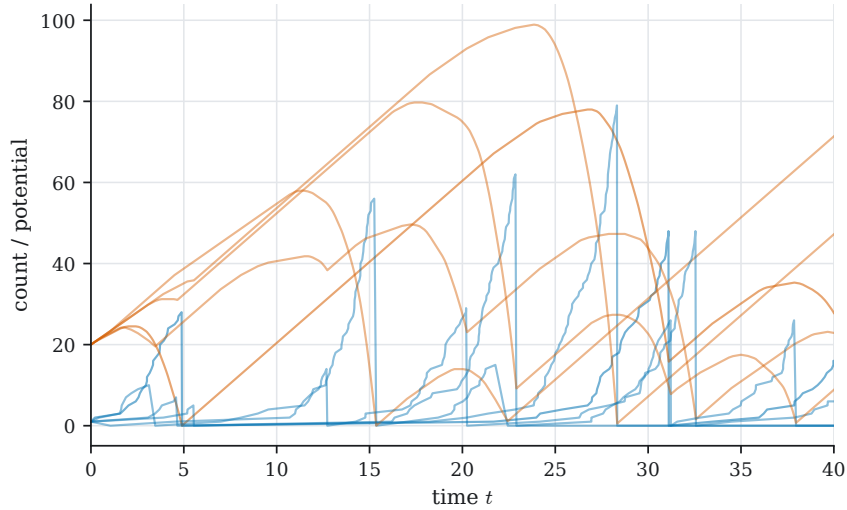


Figure 1.9: Several realisations from identical parameters and identical initial conditions. The first cycle is nearly common to all of them; later cycles are not, each catastrophe having reset the phase by a random amount. The parameters fix the amplitude, not the timing.

2. **A quasi-stationary law.** Conditional on no catastrophe having occurred, does X_t have a quasi-stationary law? For the constant-rate birth–death–catastrophe process the answer is known and geometric (????), and the general theory of extinction times for catastrophe processes is available [16, 17]. Whether the state-dependent rate $\delta X_t/V(t)$ preserves that structure is the natural first question to ask of this model, and the one most likely to have an answer.
3. **What it could be fitted to.** The model has never been fitted to anything, and what data would identify ϕ , c and δ separately is not obvious. A time series of X_t alone almost certainly would not, V being unobserved and entering only through the ratio; the question is what second observable would suffice.

1.5 The birth–plague model

1.5.1 The model

The scarcity mechanism of the previous section drove catastrophe by a potential V carried alongside a single count. This model drives the loss rate by a second count instead, so that what accelerates the loss of one component is the size of the other. Let X_t count uninfected individuals and Y_t the infected, and let the transitions in a vanishing interval be

$$\begin{aligned}\mathbb{P}(\Delta X_t = 1, \Delta Y_t = 0 \mid X_t, Y_t) &= \lambda X_t \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X_t = -1, \Delta Y_t = 1 \mid X_t, Y_t) &= (\delta + \chi Y_t) X_t \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X_t = 0, \Delta Y_t = -1 \mid X_t, Y_t) &= \mu Y_t \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X_t = \Delta Y_t = 0 \mid X_t, Y_t) &= 1 - [(\lambda + \delta + \chi Y_t) X_t + \mu Y_t] \Delta t + o(\Delta t).\end{aligned}\tag{1.27}$$

The middle channel is written once and carries two mechanisms superposed. An uninfected individual is lost to *corruption* spontaneously, at the constant per-capita rate δ , and to infection by contact with an already infected individual, at rate χ per uninfected–infected pair. Both produce the same jump, X down by one and Y up by one, so the only thing that distinguishes them is how their rates scale: the first is linear in X_t , the second bilinear in $X_t Y_t$. Writing them as separate lines with the same left-hand side would state the same channel twice rather than two channels once.

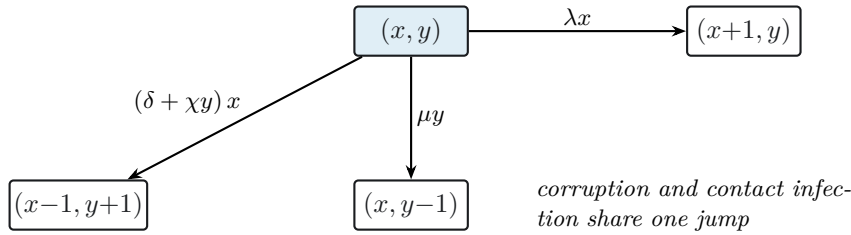


Figure 1.10: Transitions out of the state (x, y) in the birth–plague model, drawn in the convention of ???. Three channels leave each interior state, and only the diagonal one carries a rate depending on the other component — which makes it the single place where feedback enters the model: the more of the population is already infected, the faster the rest is lost. Setting $\chi = 0$ removes that dependence and leaves a linear birth process for X feeding an M/M/ ∞ -like queue, with no coupling at all.

Only one rate carries Y_t , but the coupling is genuinely two-way, because every infection removes one X and adds one Y : a large Y_t both accelerates the loss of X and is replenished by that same loss. Running against it are μY_t , which clears infected individuals, and λX_t , which replaces uninfected ones. Whether the result settles or cycles is a question about the balance of those three terms, and it is not answered here.

1.5.2 The vitality-coupled variant

The variant combines this model with that of [section 1.4](#), taking from there the idea that catastrophe — here the onset of plague rather than an extinction — is driven by scarcity per replicating agency, the agencies being those thrown off by successive bifurcations. The shape is a loose one and it fits a good many phenomena, most obviously the collapse of a population that has outgrown what sustains it.

Carry a vitality $V(t)$ alongside the pair, replenished at a constant rate and drawn down in proportion to the population:

$$\frac{dV}{dt} = \phi - c X_t.\tag{1.28}$$

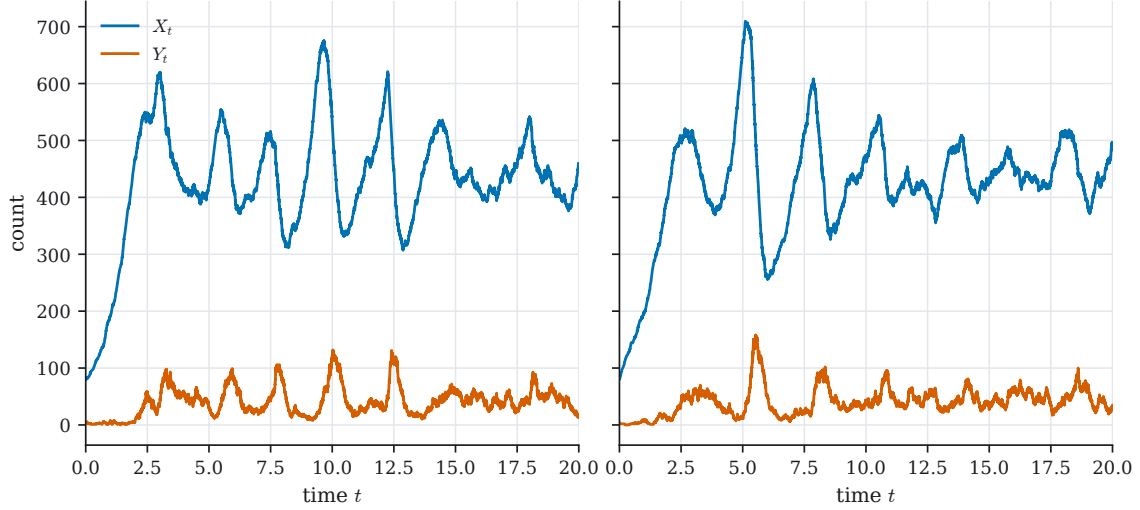


Figure 1.11: Two realisations of (1.27) at $(\lambda, \delta, \chi, \mu) = (1, 0.1, 0.02, 10)$: uninfected count X_t in blue, infected count Y_t in vermillion. The clearance rate is large relative to the contact rate, so infected individuals never accumulate and the epidemic acts as a brake on X_t rather than as a collapse of it. Set beside figure 1.9, the fluctuations never reset a phase.

The transitions are then those of (1.27) with the spontaneous part of the loss channel divided by the vitality:

$$\mathbb{P}(\Delta X_t = -1, \Delta Y_t = 1 \mid X_t, Y_t) = \left(\frac{\delta}{V(t)} + \chi Y_t \right) X_t \Delta t + o(\Delta t), \quad (1.29)$$

the other three channels being unchanged.

The reason for wanting the variant is that (1.27) has no mechanism for a collapse: its infected count rises when X_t is large and falls when it is small, which damps rather than amplifies. Dividing the corruption rate by a vitality the population itself depletes supplies the amplification that is missing, since a population which has grown large drives V down and so raises its own loss rate — exactly the ingredient section 1.4 used to close its cycle. Whether the combination gives a sustained cycle or a single collapse followed by a low-level steady state is the question the variant exists to ask, and it has not been simulated.

Open questions.

1. **Extinction of X .** What is $\mathbb{P}(X_t = 0 \text{ eventually})$? Not the standard birth–death calculation of ?? : the loss rate $(\delta + \chi Y_t)X_t$ depends on the second component, and the pair (X_t, Y_t) has no product-form generating function that closes. The obvious route is a first-step analysis on the two-dimensional lattice, in the manner of ??, and it has not been attempted.
2. **Cycles or a single collapse.** Does the vitality coupling of (1.29) produce sustained cycles, and if so with what period? This is the question section 1.5.2 was written to pose rather than to settle. Settling it wants simulation first and a first-passage analysis second, and the answer will turn on whether corruption is $\delta X_t/V(t)$ as written, or $\delta X_t^2/V(t)$.
3. **The uncoupled contact term.** If contact infection is left outside the scarcity coupling, as it is in (1.29), then at large V the model reduces to (1.27) and at small V it is dominated by corruption. Whether that crossover is sharp enough to be visible in a realisation is not known.

1.6 Public good accumulation

1.6.1 Where the idea comes from

The last two models concern *Y. pestis*, and this one borrows its mechanism from work on antimicrobial resistance. That work considers populations of two types, one resistant to a drug and one susceptible, in which the resistant type produces a substance and exports it into the surrounding medium. Because the substance protects every cell it reaches and not only the ones that made it, it is a *public good*: the susceptible type benefits from the investment of the resistant one, and the resulting dynamics are not those of a straightforward competitive exclusion [18].

Something similar may hold between *Y. pestis* and the macrophage. *Y. pestis* has evolved resistance to the toxins presented within macrophages, and were that resistance executed through a substance of this kind, confinement inside the host cell would concentrate the substance rather than disperse it — an advantage available inside a cell and not outside one. This is purely speculative and may not at all be the case, but even if it is not, the dynamics may be interesting in and of themselves. Taken as a hypothesis it would further support the argument of ?? that the within-macrophage environment carries a replicative advantage, and would supply a mechanism for that advantage rather than only an inference from behaviour.

Table 1.3: Reading of the public-good model in an intracellular *Y. pestis* setting. The last three rows are the speculation; the first four are the process as written.

Model component	Biological meaning under this reading
X_t	intracellular <i>Y. pestis</i> count in one macrophage
$P(t)$	concentration of an exported protective substance
g, c	per-bacterium production, and clearance from the enclosed volume
$\lambda(t) = \lambda_0 + \alpha P(t)$	replication raised by the accumulated good
μ	killing by macrophage toxic products, treated as constant
Enclosure	concentrates P by preventing diffusion away
Public-good mechanism	resistance executed through an exported substance; not established for this organism

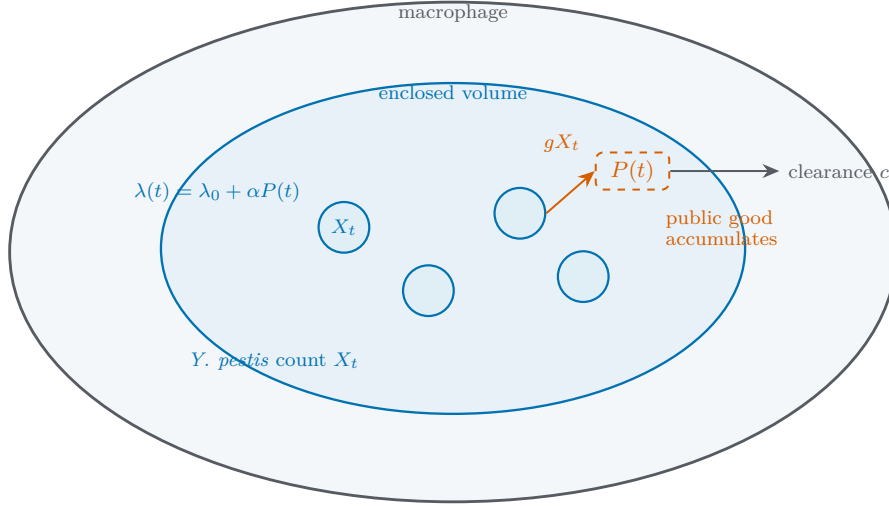


Figure 1.12: *Y. pestis* replicating inside a macrophage, with public good $P(t)$ accumulating in the enclosed volume. The enclosure stops the good diffusing away, so the same production rate gX_t buys a higher concentration inside a cell than outside; clearance c is the competing loss. Replication responds at $\lambda(t) = \lambda_0 + \alpha P(t)$.

1.6.2 The model

Let X_t count *Y. pestis* within one macrophage and $P(t)$ the concentration of public good in it. The good is produced in proportion to the bacterial count and cleared at a constant rate, and its presence raises the replication rate:

$$\begin{aligned} \frac{dP}{dt} &= gX_t - c, \\ \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda(t) X_t \Delta t + o(\Delta t), \quad \lambda(t) = \lambda_0 + \alpha P(t), \\ \mathbb{P}(\Delta X_t = -1 \mid X_t) &= \mu X_t \Delta t + o(\Delta t), \end{aligned} \tag{1.30}$$

with g the production rate per bacterium, c the clearance rate, λ_0 the base birth rate and α the public-good advantage. The effect of the host cell's toxic products is folded into the constant death rate μ , which ignores the depletion of those products that the chapter on bursting and budding pathogenesis treats explicitly; the object of interest here is the public good rather than the toxin.

The feedback is the one this chapter is about, running for once in the favourable direction: a larger population makes more of the good, which raises the birth rate, which makes the population larger still. What stops that being unconditional is the clearance term $-c$. While $X_t < c/g$ the good is cleared faster than it is made and P falls, so a small founding cohort may never start the loop at all, and the regime worth attention is therefore $\mu > \lambda_0$, where the process is subcritical on its own and turns supercritical only once enough public good has accumulated. Survival there is not a matter of luck alone but of initial numbers.

1.6.3 The minimum founding cohort

That observation can be made into a well-posed question, and it is the best one this model has. Suppose the process embedded in a host-to-host sequence: if the infection within a macrophage survives to late time the cell dies and delivers its load into two more cells, and if not, the lineage ends. This is a Galton–Watson process whose division probability is set by the within-cell

dynamics above, and it is supercritical exactly when the probability of within-cell survival exceeds one half. The quantity wanted is the smallest founding cohort achieving that:

$$K^* = \inf \left\{ K : \lim_{t \rightarrow \infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \frac{1}{2} \right\}. \quad (1.31)$$

In the regime $\mu > \lambda_0$ the limit is zero for every K once the public good is removed, so K^* is finite only on account of it and measures exactly what the public good is worth: it is the number of bacteria that must arrive together for the mechanism to pay for itself.

Figure 1.13 computes it. At $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$, so that P falls whenever $X_t < c/g = 3$, the survival probability as a function of K is a steep sigmoid through one half, and K^* itself rises with the excess death rate $\mu - \lambda_0$. The dashed line in panel (b) is c/g : K^* sits above it, as it must, because a founding cohort smaller than the clearance threshold cannot start the loop. At $\mu - \lambda_0 = 1.2$ no $K \leq 28$ crossed one half in 600 paths, which is a statement about the parameter point rather than a failure of the definition.

Two things about (1.31) are worth stating plainly. It is not a threshold in the sense of ??, which is a statement about a budget and is available in closed form; this one is defined by a limit of a survival probability, and there is no reason to expect it in closed form. The two-offspring assumption is also doing work, though less than it appears to: a general offspring law with mean m would replace the one-half in (1.31) by $1/m$, which moves K^* without changing what it measures.

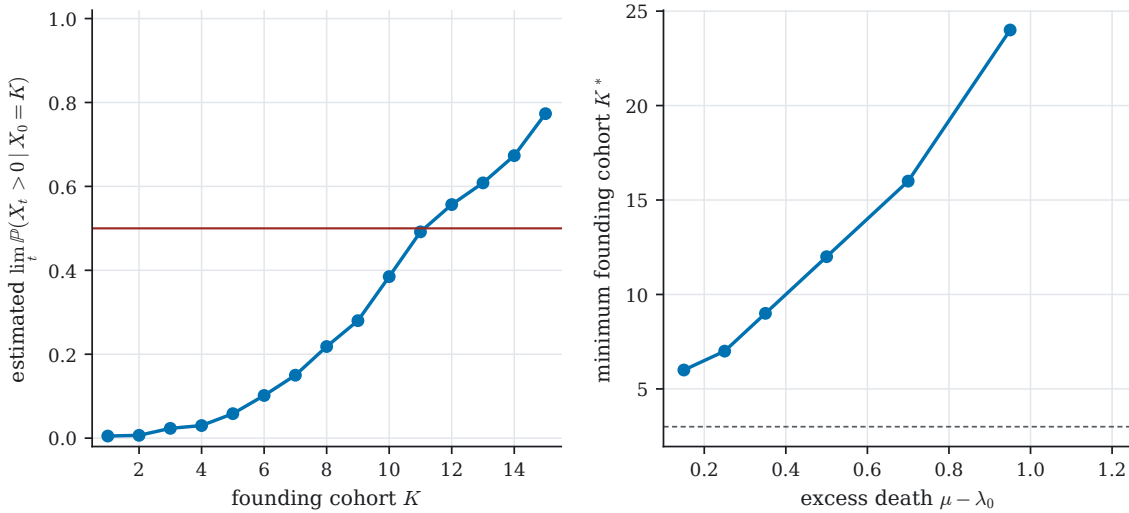


Figure 1.13: The minimum founding cohort of (1.31), computed by Gillespie simulation of (1.30) with P floored at zero (600 paths per point). Panel (a): estimated late-time survival against K at $\mu - \lambda_0 = 0.5$, with the one-half line that defines K^* . Panel (b): K^* against excess death; the dashed line is the clearance threshold $c/g = 3$. Parameters $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$.

Open questions.

1. **Death rate rather than birth rate.** It may well make more sense for the public good to lower the death rate than to raise the birth rate, since what it protects against is killing. That is a different model — $\mu(t) = \mu_0 - \alpha P(t)$ with a floor at zero, which is not a linear birth–death process in the same way — and it has yet to be written down.
2. **Whether the enclosure argument survives contact.** The whole motivation rests on confinement concentrating the good, and making that quantitative wants a volume and a diffusion or export rate, neither of which appears in (1.30). As written the model assumes the concentrating effect rather than deriving it.

3. **The scaling of K^* .** [Figure 1.13](#) is one slice of parameter space. How K^* scales with α and with c/g , and whether the rise with $\mu - \lambda_0$ remains roughly linear, is open.

1.7 The two phases of infection

In every model so far a rate has moved because the process changed its own circumstances: it used up a capacity, hardened a competitor, depleted a potential, infected its neighbours, or manufactured its own advantage. This last model is not like that. Here the rate changes because the host does. An infection proceeds for a while under one set of conditions, the host recognises it, and from that point the conditions are different ones. That difference pays for itself at once: each phase is a constant-rate process of a kind already solved, and the whole problem reduces to describing the state handed across the join. This is the section in which the mathematics closes again, and [section 1.8.1](#) takes up what exactly buys that.

1.7.1 The biology, and the shape it dictates

The pathogenesis of *Y. pestis* has two distinct phases, the pre-inflammatory and the post-inflammatory. In the first the bacterium replicates mostly intracellularly, within macrophages [\[19, 20\]](#). As the body recognises the alien presence and mounts a neutrophil response, it switches to extracellular replication [\[21, 22\]](#). That the bacterium does not simply begin extracellularly is itself the evidence that living within macrophages carries an advantage, and the simplest model of that advantage is a higher replication rate during the first phase.

The switch from intracellular to extracellular replication is coincident with the neutrophil response in the histology [\[21, 22\]](#); whether the neutrophils drive the switch, rather than arriving at the same time, is not settled by those studies and is not needed below.

Remark 1.4 (A decaying extracellular death rate). Inside the cell there are fewer of the extracellular mechanisms of innate immunity — complement, and bacteriophages. One reading, the least supported claim in this section, is that the bacterium adapts to host-specific phages within the handful of replication cycles a single infection lasts, so that the death rate in the second phase decays rather than holding constant. That would give

$$\mu_2(t) = \mu_\infty + \mu_0 e^{-t/T}, \quad (1.32)$$

with T the decay timescale, taken as the unit of time in any numerical illustration so that the exponential is e^{-t} . The two constants are independent; nothing requires the initial death rate to be twice the asymptotic one. With this rate the second phase is subcritical for a while and then not, and the size of the population handed over from the first phase is decisive. The analysed model below does not use it.

The same two-phase structure is argued from the other direction in [??](#), where the phases are distinguished by the density of the killer-cell field and the switch is placed where a removal budget ϑ^* is crossed. That account asks why the switch happens; this one asks what composing the two phases does to the probability that the infection establishes. Neither is derived from the other and they are not duplicates, but they concern one organism and one transition, and [section 1.8.4](#) returns to what reconciling them would take.

1.7.2 Phase composition

Take the first phase to be a birth–death–catastrophe process with rates $\lambda_1, \mu_1, \delta_1$ and $\lambda_1 \gg \mu_1$, so that it strongly favours replication, running until the catastrophe — the rupture of the host cell — releases its load. Take the second to be a linear birth–death process with rates λ_2, μ_2 , only slightly supercritical, so that $\lambda_2 > \mu_2$ but $\lambda_2 \approx \mu_2$. Write t_c for the switch.

The biology of [section 1.7.1](#) puts the switch at the neutrophil response, and the model puts it at the rupture of the host cell. These are treated here as one event: the rupture is what ends the intracellular phase, and the immune response is what the surrounding argument supposes provokes it. So t_c is not a fixed time chosen from outside but the catastrophe time of the first phase, and (1.33) is read conditionally on it. That is the honest reading, and it is also the one that explains why this section closes where [section 1.4](#), whose catastrophe rate is state-dependent in the same way, does not: the burst-size law ?? has already integrated the rupture time out, and hands across a distribution over $\mathcal{K}$ that no longer mentions when the rupture came.

Phase composition is then the statement that the generating function of the whole process is the second phase's, composed over the state the first phase has reached at t_c :

$$G^{\text{whole}}(x, t) = \begin{cases} G^{(1)}(x, t), & t < t_c, \\ \sum_{k \geq 0} \mathbb{P}(X_{t_c}^{(1)} = k) (G^{(2)}(x, t - t_c))^k, & t \geq t_c, \end{cases} \quad (1.33)$$

the k th power being available because the k agents handed over found independent lineages, and the weights being read off the first phase in the usual way, $\mathbb{P}(X_{t_c}^{(1)} = k) = (k!)^{-1} \partial_x^k G^{(1)}(0, t_c)$.

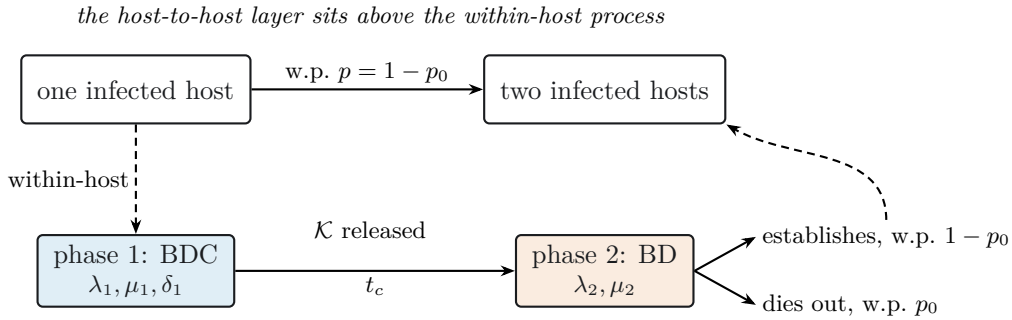


Figure 1.14: The two-phase model and the branching layer above it. Within one host the infection runs as a birth–death–catastrophe process until rupture at t_c releases a burst of size $\mathcal{K}$, and those $\mathcal{K}$ agents found independent lineages of a slightly supercritical birth–death process, which either establishes or dies out. Above it sits a Galton–Watson process with a two-point offspring law: establishment sends the infection to two new hosts, failure ends the line. The whole of the analysis below is the computation of the one number p_0 joining the two levels.

1.7.3 The host-to-host layer

Above the within-host process sits an epidemic. An infected individual either goes on to infect exactly two others or recovers and infects none, and which of the two happens is decided by whether the within-host infection establishes. Writing p for the probability that it does, the offspring count Z has

$$\mathbb{P}(Z = 2) = p, \quad \mathbb{P}(Z = 0) = 1 - p, \quad \mathbb{E}(Z) = 2p. \quad (1.34)$$

The two-point law is a modelling assumption and not a derived one. It is the crudest available caricature of transmission, and it is adopted because it makes $\mathbb{E}(Z)$ a function of p alone, p being the one quantity the within-host model supplies. Little rests on it: a general offspring law with mean m gives $\mathbb{E}(Z) = mp$, so only the multiplier moves and nothing below about how $\mathbb{E}(Z)$ responds to the phase-1 parameters is an artefact of the choice. What it does cost is the variance of Z , and with it every statement about the epidemic beyond its mean — the extinction probability of the Galton–Watson process itself depends on the whole offspring law and not merely on where it is centred.

1.7.4 The extinction probability, derived

The within-host process fails if the phase-2 population dies out, and by the law of total probability, conditioning on the burst size,

$$p_0 = \sum_{k \geq 0} \mathbb{P}(\mathcal{K} = k) \left(\frac{\mu_2}{\lambda_2} \right)^k. \quad (1.35)$$

The sum is worth naming: (1.35) is the *phase-1 burst-size generating function evaluated at the phase-2 extinction probability*, $p_0 = G_{\mathcal{K}}(\mu_2/\lambda_2)$. Both of its ingredients are proved elsewhere in this thesis, and neither is re-derived here.

The first is the burst-size law. For a birth–death–catastrophe process the load released at rupture is geometric on the larger root, $\mathbb{P}(\mathcal{K} = k) = (\delta_1/\lambda_1)a^{-k}$ for $k \geq 1$, which is ??, with a and b the roots of the characteristic quadratic of ?. The second is the extinction probability of a supercritical linear birth–death process from one founder, μ_2/λ_2 , and hence $(\mu_2/\lambda_2)^k$ from k independent founders, which is ?.

The $k = 0$ term is the mass on paths that never rupture at all: the cell’s occupants die out before the catastrophe can come, no load is released, and no onward host is infected. That mass is b , and it is not a free parameter. ? is stated unconditionally, including the paths that never burst, and its normalisation is $\sum_{k \geq 1} (\delta_1/\lambda_1)a^{-k} = (\delta_1/\lambda_1)/(a - 1) = 1 - b$, so

$$b = 1 - \frac{\delta_1/\lambda_1}{a - 1}, \quad (1.36)$$

which is also the smaller root of the same quadratic. The three constants a , b and δ_1/λ_1 are determined by $(\lambda_1, \mu_1, \delta_1)$ together and only two of them may be chosen independently.

The sum in (1.35) is geometric and closes. With $x = \mu_2/(a\lambda_2)$,

$$\sum_{k \geq 1} \frac{\delta_1}{\lambda_1} a^{-k} \left(\frac{\mu_2}{\lambda_2} \right)^k = \frac{\delta_1}{\lambda_1} \cdot \frac{x}{1 - x},$$

so that

$$p_0 = b + \frac{\delta_1}{\lambda_1} \cdot \frac{\mu_2}{a\lambda_2 - \mu_2}, \quad \text{convergent if and only if } a\lambda_2 > \mu_2, \quad (1.37)$$

and therefore

$$\mathbb{E}(Z) = 2 \left(1 - b - \frac{\delta_1}{\lambda_1} \cdot \frac{\mu_2}{a\lambda_2 - \mu_2} \right). \quad (1.38)$$

The convergence condition costs nothing: the second phase is supercritical by assumption, so $\lambda_2 > \mu_2$, and $a > 1$ by ?, so $a\lambda_2 > \lambda_2 > \mu_2$ always.

What (1.37) buys is a dependence of the epidemic on the intracellular phase that is explicit rather than numerical. The first phase appears in (1.38) only through a and δ_1/λ_1 , and both fall as δ_1 falls, a cell less inclined to rupture holding its occupants longer and releasing more of them. The second enters only through the ratio μ_2/λ_2 , and the product $a\lambda_2$ in the denominator is the whole of the interaction between them, and it is where the leverage is: a larger burst multiplies the founding cohort of a near-critical process, and since extinction from k founders is the k th power of a number close to one, multiplying k is the most effective thing that can be done to it.

1.7.5 Other compositions

The composition above is the birth–death–catastrophe to birth–death case. Three nearby variants share the same two-phase skeleton and differ only in which parameter does the work: a first phase that is an ordinary birth–death process running for a fixed time; the BDC case just analysed, in which δ_1 sets duration and handover together; and a first phase that simply

delivers a fixed count, the comparator against which the others are read. A fourth variant does not share the machinery. It is a fixed handover into the decaying death rate of [remark 1.4](#), and with μ_2 a function of time the extinction probability of the second phase is no longer μ_2/λ_2 but the solution of the backward Riccati equation ???. It is named here and left there.

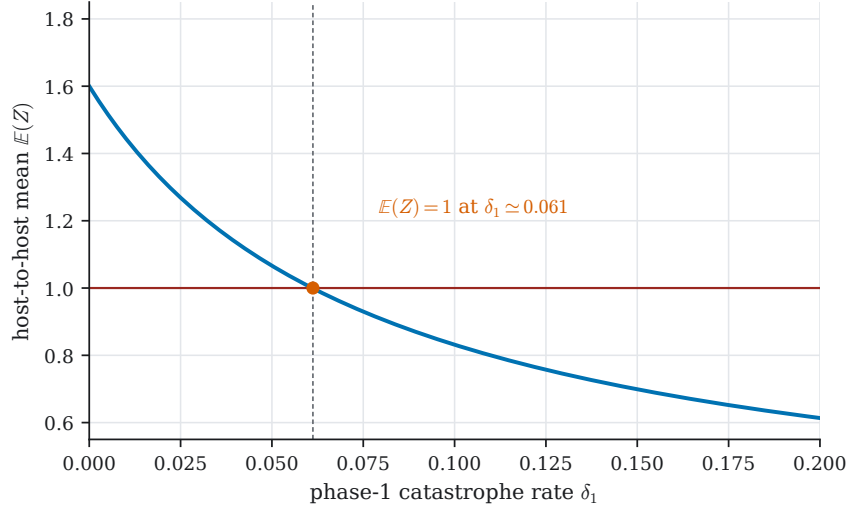


Figure 1.15: The host-to-host mean $\mathbb{E}(Z)$ against the phase-1 catastrophe rate δ_1 , from the closed form (1.38), at $(\lambda_1, \mu_1) = (1, 0.2)$ and $(\lambda_2, \mu_2) = (1, 0.9)$. The second phase is barely supercritical, so a lineage founded by a single agent almost certainly fails; what rescues the infection is arriving in numbers. A small δ_1 is a longer first phase and a larger release, and that is what carries $\mathbb{E}(Z)$ above one. The unit crossing is at $\delta_1 \simeq 0.061$.

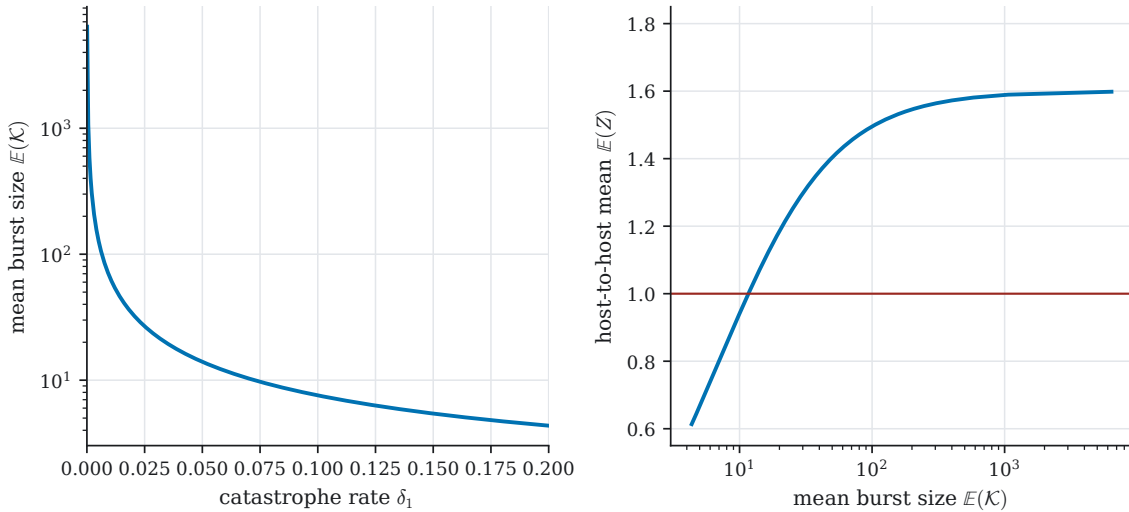


Figure 1.16: The same dependence, decomposed. Panel (a): mean burst size against δ_1 , a lower propensity to burst yielding a larger burst. Panel (b): $\mathbb{E}(Z)$ against mean burst size, a larger founding cohort raising the chance of surviving the near-critical extracellular phase. Read together, δ_1 acts on the epidemic entirely through the burst size, which is what (1.38) makes explicit.

Open questions.

1. **The decaying-rate variant.** The decaying- $\mu_2(t)$ model of [remark 1.4](#) is named and not analysed. Taking it up needs the inhomogeneous extinction probability of ?? in place of μ_2/λ_2 .
2. **Against the threshold budget.** ?? reaches the same organism and the same transition with different machinery, and offers a threshold budget ϑ^* at which the intracellular strategy stops paying. How $\mathbb{E}(Z)$ here compares against that threshold — whether the δ_1 that puts $\mathbb{E}(Z)$ above one is the one that puts the switch where ϑ^* puts it — is the question that would join the two accounts.

1.8 Discussion

1.8.1 What the collection shows

The sorting of [figure 1.1](#) is the argument, and it is worth being exact about the last cell. A rate that moves because the process moved costs the generating function, and with it most of what the preceding chapters were able to do — unless what the process acts on is a single random time rather than a moving rate. Between those two poles, the interesting object in the collection is the Pimentel chain: a process that repeatedly restores its own criticality is not the same thing as one that approaches an equilibrium, and [figure 1.6](#) is the picture of that difference.

[Section 1.7](#) sits where it does because the phase ends at the rupture of the host cell, which the process itself brings on at rate $\delta_1 X_t$. That is feedback of the same kind as [section 1.4](#)’s. What differs is where the feedback is allowed to show up. In every other model the state bends a rate, and the generating function stops closing. Here it sets a single random *time*, and the burst-size law has already integrated that time out — what crosses the join is a distribution over $\mathcal{K}$ that has forgotten when the rupture came.

That is the trade the chapter makes visible, and it is sharper than feedback against no feedback. A process may act on its own rates as much as it likes and still be solvable, provided what survives the interaction is a random time or a random count rather than a moving rate. It is continuous bending, not feedback as such, that costs the generating function.

1.8.2 Limitations

Five of the seven models have never been compared against data, and for three of those it is not obvious what data would serve. The Pimentel+ specification of [section 1.3.6](#) has not been simulated, so the behaviour attributed to it is what the construction was intended to produce rather than anything observed of it. The public-good threshold K^* of [figure 1.13](#) is a numerical estimate on one parameter slice, not a comparison against data.

The ecological models of [sections 1.2](#) and [1.4](#) use “species” as a unit whose delimitation is contested in the literature the speciation section responds to. [Section 1.4](#) is explicit that its X_t might equally count individuals, species or organisational units.

None of [sections 1.3.6](#) and [1.4](#) to [1.6](#) carries an existence result for the behaviour its motivation suggests. That the vitality coupling gives a sustained oscillation rather than one collapse, and that the fitness penalty of (1.24) can reverse a dominance, remain conjectures supported by the construction.

The organism-level claims of [sections 1.6](#) and [1.7](#) are hypotheses about *Y. pestis* compatible with the cited intracellular literature, not inferences this chapter can check.

1.8.3 Open problems

1. **The mixture at β_5 .** The law of $\mathcal{B}(t)$ under a randomly switching capacity, and the resulting mixture of geometrics (1.15). Tractable: a Markov-modulated integral.
2. **Identifiability.** What a time series of S_t identifies, given that only $\mathcal{B}$ enters the law, and whether β_1 and β_3 can be told apart from one.
3. **Whether the Pimentel oscillation persists.** Every realisation run here eventually fixated. Whether the expected number of reversals before fixation grows without bound in α , ε or N , or saturates, is the question the model was built to answer.
4. **A diffusion limit** in which (1.20) is a moving unstable equilibrium, so that the divergence of nearby trajectories becomes a rate.
5. **The scaling of K^* .** Figure 1.13 is one slice. How the threshold moves with α and c/g is open.
6. **Against the threshold budget.** How $\mathbb{E}(Z)$ here compares against ϑ^* of ???: whether the δ_1 that puts $\mathbb{E}(Z)$ above one is the δ_1 that puts the switch where the removal budget puts it.

Of these, the mixture and the ϑ^* comparison would repay attention first: each is well posed, and each closes a section that currently stops one step short of an answer.

1.8.4 Connections

?? identifies a selection gradient between release phenotypes and hands the study of variability itself to this chapter. Sections 1.2, 1.3 and 1.3.6 are what discharges that: variability as the object rather than as a property of a fixed mechanism.

Two of the models point back into the core theory. The quasi-stationary question of section 1.4 is ??’s own question asked of a state-dependent catastrophe rate; the extinction probability of section 1.5 wants the first-step analysis ?? runs on a two-dimensional lattice, applied to a lattice of a different shape.

The two *Y. pestis* models point forward. Section 1.6 offers a mechanism for the intracellular advantage that the chapter on bursting and budding pathogenesis infers but does not explain, and section 1.7 offers $\mathbb{E}(Z)$ for the consequence of the two-phase strategy at the level of an epidemic. The sixth open problem above is what would join them to it.

Model	State	Rates or evolution equation	Parameters	Figures	Status
Logistic speciation section 1.2	$S_t \in \{1, 2, \dots\}$, $N(t) \in (0, K)$	$dN/dt = \varrho N(1 - N/K)$; $q_{s,s+1}(t) = \beta(t)s$ with $\beta(t) = \sigma(K - N(t))$	ϱ growth; K capacity; N_0 founding abundance; σ speciation coefficient	figures 1.2 and 1.3	closed form
Pimentel competition section 1.3	$(A_t, R_{a,t}, R_{b,t})$, with $A_t \in \{0, \dots, N\}$, discrete time	kernel (1.16) with blocking (1.17); $R_{k,t+1} = R_{k,t} + \varepsilon \Delta R_{k,t}$, (1.19)	N capacity; ε evolution rate; α squeeze exponent	figures 1.4 to 1.7	critical point closed form; simulated specified
Pimentel+ section 1.3.6	$(A_t, f_a, f_b, S_t^{(a)}, S_t^{(b)})$	kernel (1.21); fitness (1.24); species counts (1.23)	ε, α as above; ζ dominance exponent; γ variation penalty; μ_a species extinction	—	—
Rise, run, ruin, rejuvenation section 1.4	$(X_t, V(t))$, $X_t \in \{0, 1, \dots\}$	$dV/dt = \phi - cX_t$; channels (1.26), catastrophe at $\delta X_t/V(t)$	ϕ inflow; c consumption; λ proliferation; δ catastrophe; ω revival	figures 1.8 and 1.9	simulated
Birth–plague section 1.5	$(X_t, Y_t) \in \mathbb{Z}_{\geq 0}^2$	(1.27); one loss channel at $(\delta + \chi Y_t)X_t$	λ birth; δ corruption; χ contact; μ clearance	figures 1.10 and 1.11	simulated
Birth–plague, vitality-coupled section 1.5.2	$(X_t, Y_t, V(t))$	as above with corruption at $\delta X_t/V(t)$, (1.29); $dV/dt = \phi - cX_t$	as above, plus ϕ inflow and c consumption	—	specified
Public good section 1.6	$(X_t, P(t))$	$dP/dt = gX_t - c$; birth at $(\lambda_0 + \alpha P(t))X_t$, death at μX_t , (1.30)	g production; c clearance; λ_0 base birth; α advantage; μ death	figures 1.12 and 1.13	K^* computed
Two phases of infection section 1.7	$X_t^{(1)}$ then $X_t^{(2)}$; Z above	BDC $(\lambda_1, \mu_1, \delta_1)$ to t_c , then BD (λ_2, μ_2) from $\mathcal{K}$ founders; composition (1.33)	$\lambda_1, \mu_1, \delta_1$; λ_2, μ_2 ; t_c switch time	figures 1.14 to 1.16	closed form

Table 1.4: Every model in the chapter, with what it acts on, what moves its rates, and how far it was taken. The rows run in section order, so the column of statuses reads as the shape of the chapter. Three constructions recur across otherwise unrelated rows and are worth noticing: an inflow ϕ set against a per-capita consumption c , in three models; the exponent structure $(\cdot)^\alpha$ near an absorbing boundary, in two; and a rate written as a base plus a coupling, in three.

APPENDICES

1.A Model summary

1.B Figure and simulation record

Line plots in this chapter are vector PDFs, drawn in the house palette of section 1.1’s sibling chapters, from either a closed form or a recorded-seed simulation of the equations as printed. Two schematics remain inline TikZ; two more are standalone TikZ packages. The spatial automaton is the one remaining raster, cropped from the original frames. The rise–run–ruin–rejuvenation panels figures 1.8 and 1.9 were re-simulated at the invented tuple $(\phi, c, \lambda, \delta, \omega) = (4, 0.8, 0.5, 0.25, 0.15)$, because the parameters of the original rasters were never recorded: the cycle is the same, the amplitudes are not.

Figure	File	Parameters / seed	Code
figure 1.1	fig01_taxonomy/fig01.pdf	schematic	TikZ
figure 1.3	fig02_logspec_mean/fig02.pdf	$K = 10, \varrho = 1, \sigma = 0.08;$ $N_0 = 1, 2, 5$	closed form
figure 1.2	fig02b_logspec_limit/fig02b.pdf	$K/N_0 \in \{10, 50\},$ $\sigma K/\varrho \in \{1, 2\};$ (1.12)	closed form
figure 1.4	fig03_pimentel_runs/fig03.pdf	$\alpha = 1, N = 100, \varepsilon = 10;$ seeds 19, 3	discrete chain
figure 1.5	fig04_pimentel_ic/fig04.pdf	$\alpha = 10, N = 10^5, \varepsilon = 1, 2 \times 10^6$ steps; seeds 4, 7	discrete chain
figure 1.6	fig05_pimentel_icrit/fig05.pdf	$\alpha = 3, N = 10^4, \varepsilon = 1;$ seed 3	discrete chain
figure 1.7	fig06_automaton/fig06.jpg	six cropped frames of the C++/SFML automaton	raster
figure 1.8	fig07_rrrr_cycles/fig07a.pdf	$\phi = 4, c = 0.8, \lambda = 0.5,$ $\delta = 0.25, \omega = 0.15;$ seeds 3, 11	Gillespie
figure 1.9	fig07_rrrr_cycles/fig07b.pdf	as above; seeds 3, 5, 8, 11, 17	Gillespie
figure 1.10	inline TikZ	schematic	in source
figure 1.11	fig09_plague_paths/fig09.pdf	$(\lambda, \delta, \chi, \mu) = (1, 0.1, 0.02, 10);$ seeds 2, 8	Gillespie
figure 1.12	fig10_public_good/fig10.pdf	schematic	TikZ
figure 1.13	fig14_kstar/fig14.pdf	$(\lambda_0, \alpha, g, c) =$ $(0.6, 0.25, 0.15, 0.45);$ 600 paths	Gillespie
figure 1.14	inline TikZ	schematic	in source
figure 1.15	fig12_EZ/fig12.pdf	$(\lambda_1, \mu_1) = (1, 0.2),$ $(\lambda_2, \mu_2) = (1, 0.9)$	(1.38)
figure 1.16	fig13_EZ_decomp/fig13.pdf	as above	(1.38)

Table 1.5: Provenance of every figure. Scripts live in `figures/figXX_*/src/`; the shared palette is `figures/_style/style_rc.py`. Seeds are those that produced the shipped files.

1.C The inhomogeneous Yule generating function

The proof of [proposition 1.2](#) is the standard generating-function computation for a pure birth process, with βt replaced by $\mathcal{B}(t)$. It is recorded here rather than in the body, because the method of characteristics is the mathematical introduction's and is not being rebuilt.

1.C.1 The generating function

The forward Kolmogorov equation for this system is that of the pure birth process with the rate carrying a t ; writing $p_n(t) = \mathbb{P}(S_t = n)$ and expanding over a vanishing interval gives

$$\frac{dp_n}{dt} = \beta(t)(n-1)p_{n-1} - \beta(t)n p_n, \quad (1.39)$$

an instance of [??](#). Multiplying by z^n and summing turns it into a first-order partial differential equation for $G(z, t) = \sum_{n \geq 1} p_n(t) z^n$,

$$\frac{\partial G}{\partial t} = \beta(t) z(z-1) \frac{\partial G}{\partial z}, \quad (1.40)$$

and despite the time dependence this still yields to the method of characteristics, [??](#). Introduce characteristic variables (s, τ) with

$$z(s, \tau = 0) = s, \quad t(s, \tau = 0) = 0, \quad (1.41)$$

along which

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dz}{d\tau} = \beta(t) z(z-1). \quad (1.42)$$

The first says G is a function of s alone and the second gives $t = -\tau$. With a single founding species $G(z, 0) = z$, so on the initial curve $G = s$, and therefore $G = s$ everywhere: the whole problem is to express s in terms of z and t .

The third characteristic equation separates. Using $d\tau = -dt$ on the right,

$$\int \frac{dz}{z(z-1)} = \int \beta(-\tau) d\tau = - \int \beta(t) dt, \quad (1.43)$$

and since the left side integrates to $\ln((z-1)/z)$ this is

$$\ln \left(\frac{z-1}{z} \right) = -\mathcal{B}(t) + \text{const.}$$

The initial conditions (1.41) fix the constant as $\ln((s-1)/s)$, because $\mathcal{B}(0) = 0$. Hence $(z-1)/z = ((s-1)/s)e^{-\mathcal{B}(t)}$, which rearranges to

$$s = \left(1 - \frac{z-1}{z} e^{\mathcal{B}(t)} \right)^{-1},$$

and so

$$G(z, t) = \frac{z}{z + (1-z)e^{\mathcal{B}(t)}}. \quad (1.44)$$

Differentiating with respect to z and setting $z = 1$ returns $\mathbb{E}(S_t) = e^{\mathcal{B}(t)}$, which is (1.8) again by a different route.

1.C.2 The distribution

Write $p = e^{-\mathcal{B}(t)}$ and divide numerator and denominator of (1.44) by $e^{\mathcal{B}(t)}$:

$$G(z, t) = \frac{pz}{1 - (1-p)z}, \quad (1.45)$$

which is the probability generating function of a geometric law on $\{1, 2, \dots\}$ with success parameter p . That is (1.7), and it completes the proof of proposition 1.2. Repeated differentiation of (1.44), which was the route originally taken, gives $\partial_z^n G = n! e^{\mathcal{B}} (e^{\mathcal{B}} - 1)^{n-1} (z(1-e^{\mathcal{B}}) + e^{\mathcal{B}})^{-(n+1)}$, and the inversion formula $\mathbb{P}(S_t = n) = (n!)^{-1} \partial_z^n G|_{z=0}$ of ?? returns the same law.

Bibliography

- [1] Ronald Aylmer Fisher. *The genetical theory of natural selection: a complete variorum edition*. Oxford University Press, 1929.
- [2] Friedrich Nietzsche. Beyond good and evil. In *Moral Disagreements*, pages 81–88. Routledge, 2013.
- [3] Niles Eldredge and Stephen Jay Gould. Punctuated equilibria: An alternative to phyletic gradualism. In Thomas J. M. Schopf, editor, *Models in Paleobiology*, pages 82–115. Freeman, Cooper, San Francisco, 1972.
- [4] Hélène Morlon. Phylogenetic approaches for studying diversification. *Ecology Letters*, 17(4):508–525, 2014.
- [5] Sean Nee, Robert M. May, and Paul H. Harvey. The reconstructed evolutionary process. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 344:305–311, 1994.
- [6] Graham E Budd and Richard P Mann. History is written by the victors: the effect of the push of the past on the fossil record. *Evolution*, 72(11):2276–2291, 2018.
- [7] Per Bak and Kan Chen. Self-organized criticality. *Scientific American*, 264(1):46–53, 1991.
- [8] Benjamin Gompertz. XXIV. On the nature of the function expressive of the law of human mortality, and on a new mode of determining the value of life contingencies. In a letter to Francis Baily, Esq. FRS &c. *Philosophical transactions of the Royal Society of London*, 115:513–583, 1825.
- [9] George U. Yule. A mathematical theory of evolution, based on the conclusions of Dr. J. C. Willis, F.R.S. *Philosophical Transactions of the Royal Society of London. Series B, Containing Papers of a Biological Character*, 213:21–87, 1925.
- [10] David Pimentel, Edwin H Feinberg, Peter W Wood, and John T Hayes. Selection, spatial distribution, and the coexistence of competing fly species. *The American Naturalist*, 99(905):97–109, 1965.
- [11] G. Evelyn Hutchinson. *The Ecological Theater and the Evolutionary Play*. Yale University Press, New Haven, 1965.
- [12] Craig M Pease. On the evolutionary reversal of competitive dominance. *Evolution*, 38(5):1099–1115, 1984.
- [13] Ilya Prigogine and Isabelle Stengers. *Order out of chaos: Man's new dialogue with nature*. Verso Books, 2018.
- [14] Avshalom C Elitzur. Let there be life: Thermodynamic reflections on biogenesis and evolution. *Journal of theoretical biology*, 168(4):429–459, 1994.
- [15] Abhishek Sharma, Dániel Czégel, Michael Lachmann, Christopher P Kempes, Sara I Walker, and Leroy Cronin. Assembly theory explains and quantifies selection and evolution. *Nature*, 622(7982):321–328, 2023.
- [16] Peter J Brockwell. The extinction time of a birth, death and catastrophe process and of a related diffusion model. *Advances in Applied Probability*, 17(1):42–52, 1985.
- [17] Ben Cairns and PK Pollett. Extinction times for a general birth, death and catastrophe process. *Journal of applied probability*, 41(4):1211–1218, 2004.
- [18] Matthew Asker, Lluís Hernández-Navarro, Alastair M Rucklidge, and Mauro Mobilia. Coexistence of competing microbial strains under twofold environmental variability and demographic fluctuations. *New Journal of Physics*, 25(12):123010, 2023.
- [19] Céline Pujol and James B. Bliska. The ability to replicate in macrophages is conserved between *Yersinia pestis* and *Yersinia pseudotuberculosis*. *Infection and Immunity*, 71(10):5892–5899, 2003.
- [20] Michael G. Connor, Amanda R. Pulsifer, Donghoon Chung, Eric C. Rouchka, Brian K. Ceresa, and Matthew B. Lawrenz. *Yersinia pestis* targets the host endosome recycling pathway during the biogenesis of the *Yersinia*-containing vacuole to avoid killing by macrophages. *mBio*, 9(1):e01800–17, 2018.
- [21] Catharine M. Bosio, Andrew W. Goodyear, and Steven W. Dow. Early interaction of *Yersinia pestis* with APCs in the lung. *The Journal of Immunology*, 175(10):6750–6757, 2005.
- [22] Kristen N. Peters, Miqdad O. Dhariwala, Jennifer M. Hughes Hanks, Corrie R. Brown, and Deborah M. Anderson. Early apoptosis of macrophages modulated by injection of *Yersinia pestis* YopK promotes progression of primary pneumonic plague. *PLoS Pathogens*, 9(4):e1003324, 2013.